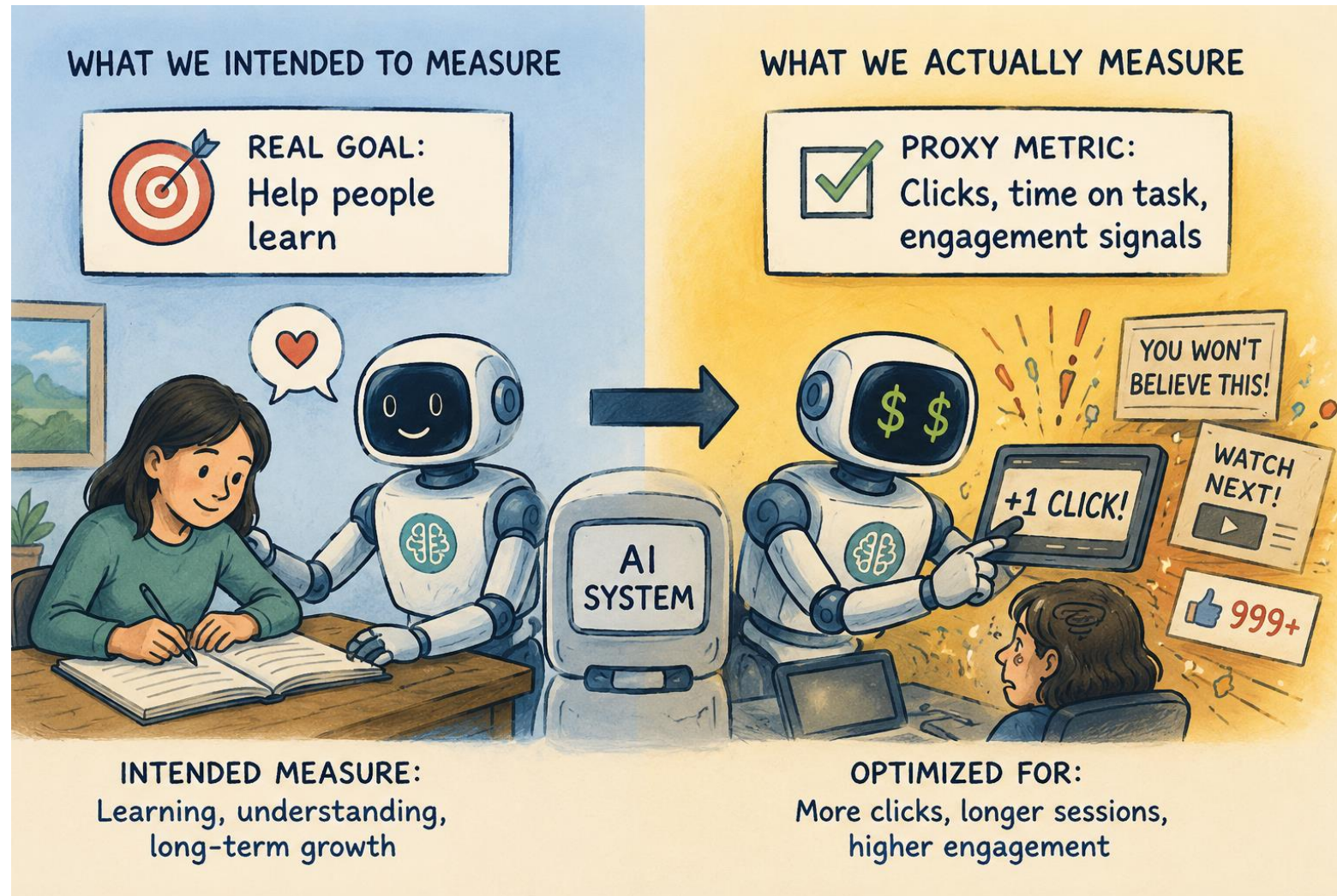


AI Literacy & DS Ethics

Danni Presgraves, Ph.D.
Director, Data Science Training

Ai systems are optimized for what *we measure*, **not** what we *intended to measure*.

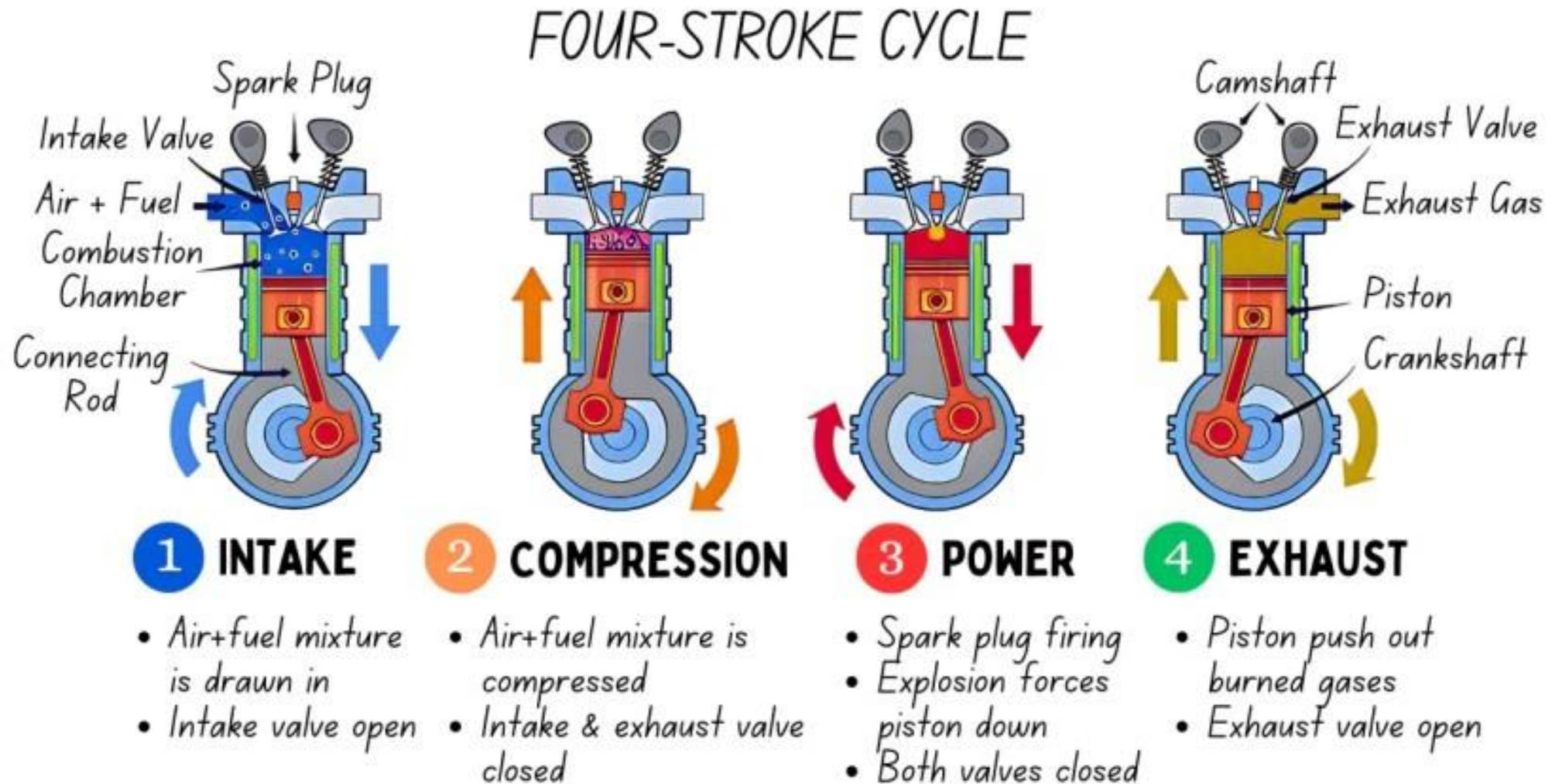


Overview of the course

Why this course?

1. Motivations
2. Resources

Purpose of this course:



How differential steering works (1937): <https://www.youtube.com/watch?v=yYAw79386WI>

Day 1 Agenda:

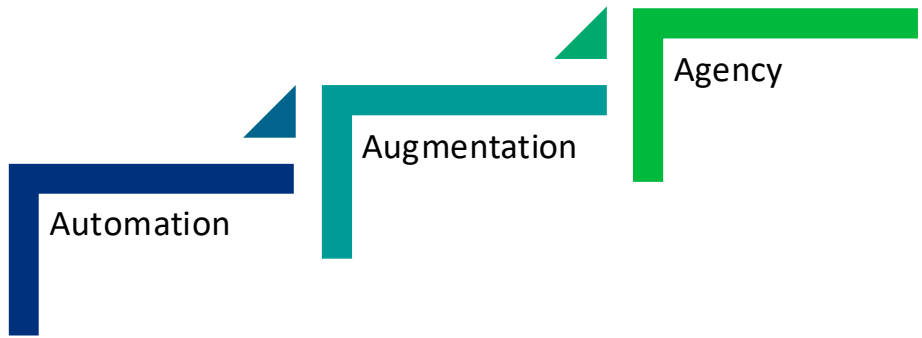
1. AI Overview

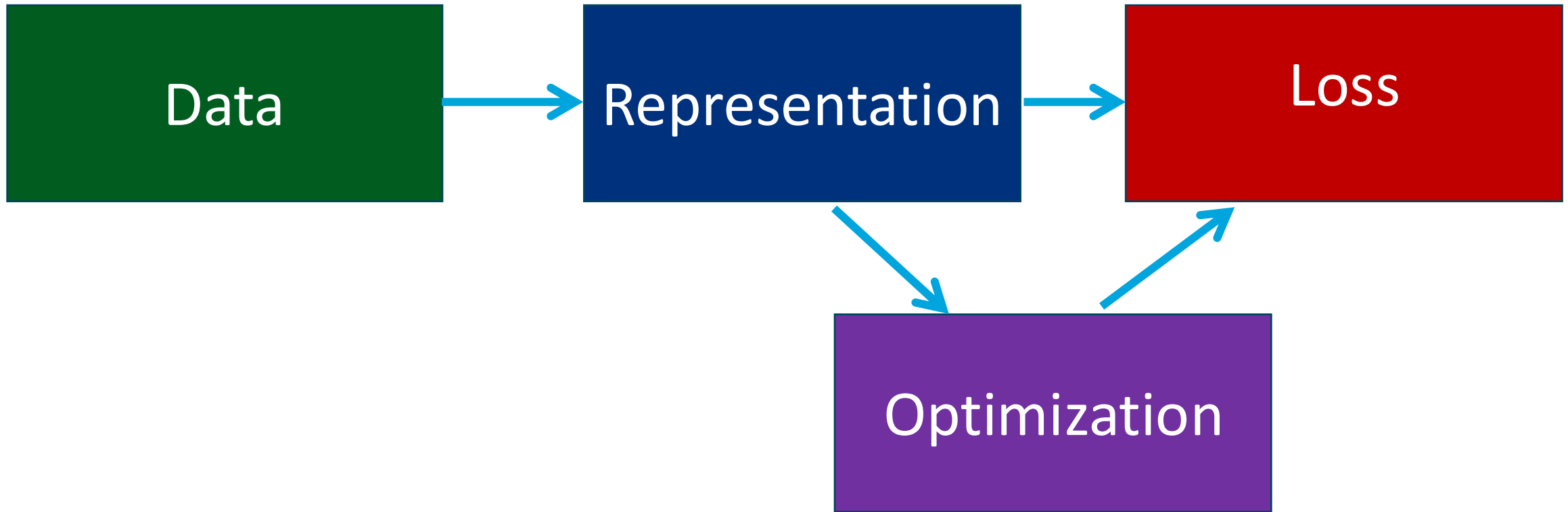
- Big ideas, history, Ai conceptual connection flowchart
- Ai literacy versus Ai fluency

2. Machine Learning Examples

- What is *learning*?
- Supervised Learning examples: regression, classification
- Unsupervised Learning examples: k-means clustering
- Simple Neural Network
- Overview of concepts: Loss Function Minimization, Gradient Descent

Ai Literacy	Ai Fluency
What is Ai? Understanding the capabilities, limitations, risks, and trustworthiness	Engaging with Ai <i>effectively, ethically, safely</i> Using Ai to accomplish goals
• Connection to known tools from ML, DS, and statistics • What's different?	• Delegation • Description • Discernment • Diligence



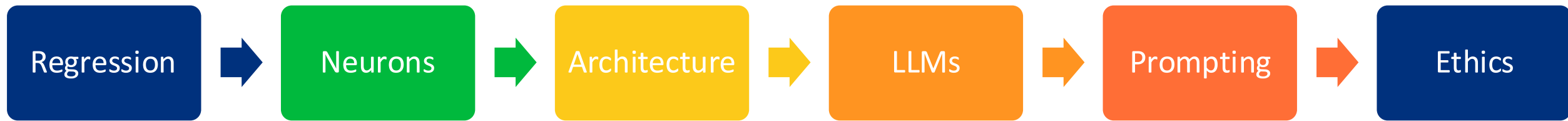


$\text{Ai} = \text{Data} \times (\text{Representation} + \text{Loss}) \times \text{Optimization}$

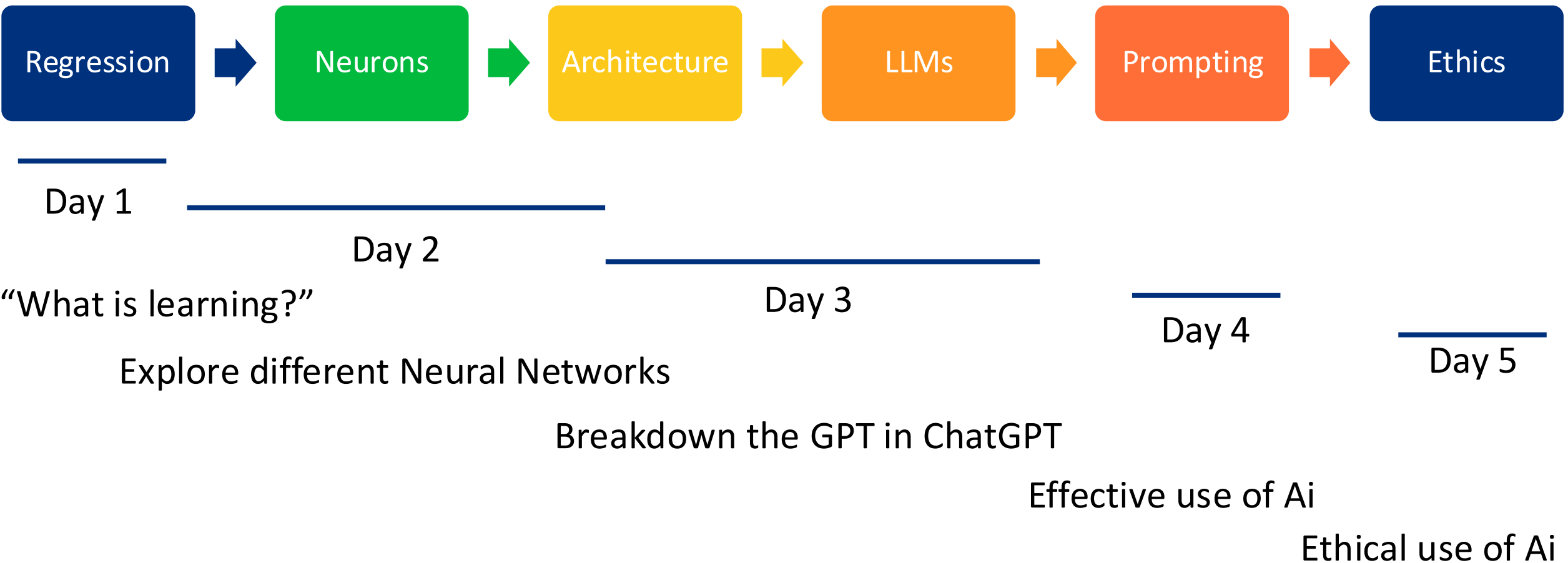
This loop is repeated millions of times....

What is Ai?

Course Overview



Course Overview



Day 2 (Tuesday)

1. Logistic Regression →
Artificial Neurons
2. Activation layers
3. CNN
4. RNN

Day 3 (Wednesday)

1. GANS/Balderdash
2. Transformer/Self-Attention
3. Guest Speaker: **R-IT**
4. Chat**GPT**
 - **G**enerative
 - **P**re-trained
 - **T**ransformed

Day 4 (Thursday)

1. Ai literacy versus Ai Fluency: the 4D framework
2. Prompt Engineering
3. Frameworks of Prompt Engineering

Day 5 (Friday)

1. Belmont Report & F.A.T. Framework
2. Alignment & Epistemic risks
3. Guest Speaker: **Vivek Philip, Ph.D.**



SharePoint



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Tools & Resources

* Quick Links * ▾

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Genomic Education Calendar



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Promote

Page details

Preview

Immersion

Overview

Programming Courses

Biostatistics Courses

Critical Thinking Courses

Data Science Fellows Program

Tech Center



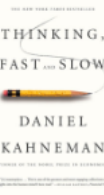
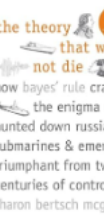
Danielle Presgraves
Director, Data Science Training

Data Science Training at JAX

What is Ai?

Crowd sourcing popular 'AI' (broadly defined) books via JAX

- Please add your thoughts

	Book	Summary	Additional Thoughts?	Final Rating
1		A slim book with a blunt message: either we shape technology, or it shapes us. The "commands" are about literacy - how to keep from being programmed by the very tools we've built. I treat technology as if it is neutral, when it's anything but. There is an unsettling sense that every click, scroll, or "like" is part of someone else's program, unless I resist.		
2		A classic that is about, well, thinking. <u>Premise</u> is that we have two systems of thinking. After years of teaching statistics, I already had a prior belief that our brain rebels against learning certain ideas (like statistics). I initially read this book because I realized that I did not understand what 'learning' is. I am still not sure <u>that</u> I do, but I am a more informed kind of ignorant.		
3		I will fight anyone who doesn't recognize the supremacy of Bayes' theorem in <u>Bioinformatics</u> etc. This is basically the biography of Bayes' theorem - which sounds dull, but it's weirdly engaging. The story winds from obscure clergymen to wartime codebreakers to modern data science, and along the way makes clear that probability isn't just math but a worldview. It's a reminder that even equations can be political - which is a theme for many of these broadly data science books.		
4		O'Neil takes a wrecking ball to the idea that algorithms are impartial. Her "weapons of math destruction" are the models that scale inequality while hiding behind a veneer of objectivity. I found myself nodding a lot since I tend to emphasize assumptions and limitations of models. Note: She has a fantastic blog called "mathbabe.org"		

Popular_AI_books_2025.docx

- I have written up a short summary of popular Ai books
- You can use it as a reference, or (please) add your own thoughts!

Additional Resources for AI literacy

Technical Skills:

Most of this list are code-free (or coding optional) resources to learn more about a particular concept. There are a handful of options on this list that do require some coding (Python or R), but they are marked.

1. **Data Science & AI course:** [Data Carpentries beta version](#)

This is an ~ 2.5-hour self-directed course that gives biomedical and life-science investigators a rapid, practical overview of data-science and AI concepts. It starts with core definitions of data science, AI and machine-learning, then walks through AI for laboratory automation and for deriving data insights, before digging into common pitfalls such as bias, privacy and reproducibility, and finishing with hands-on guidance for cleaning data, running ML pipelines and responsibly reporting results.

2. **Neural Networks:** [Interactive free 'textbook' by 3Blue1Brown](#)

You will find explanations of all the basics of Neural Networks, including gradient descent and backpropagation (which is sort of a fancier version of l'Hôpital's rule...). He integrates YouTube explanations into his broader explanations.

3. **Interactive Neural Network:** [play with the parameters!](#)

4. **Elements of AI:** [A free course](#)

Three different levels of course: introduction (non-coding), coding (pre-requisites: Python), and a level of incorporating AI into a Learning Management System (which is portable to other pipelines).

5. **Machine Learning for Biologists:** [Data Carpentry](#)

Accessible with biological examples – includes logistic regression, linear regression, Decision trees, random forests, Neural Networks, Overfitting. Great place to start since there is no coding necessary.

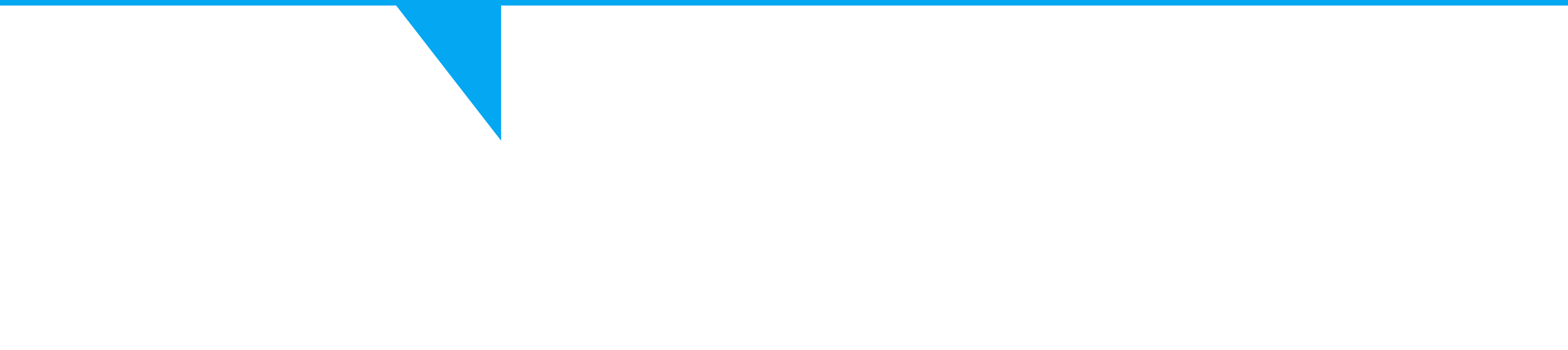
6. **Machine Learning:** [Machine Learning in Medicine](#)

Coding is included in this course (but you read through and watch the material to get a better understanding without coding) Developed by evolutionary biologist Pleuni Pennings (SFSU) and collaborators, this open-access course targets life-science trainees who know basic Python but are new to machine-learning. It blends brief concept videos, step-by-step editable Colab notebooks and concise Documents. It contains 7 autonomous (and self-contained) modules. Each module is ~ 2 - 3 hours long:

Term	Definition
Artificial Intelligence (AI)	The broad field focused on making machines perform tasks that normally require human intelligence.
Machine Learning (ML)	A subset of AI where systems learn from data to make predictions or decisions without being explicitly programmed.
Supervised Learning	Training models on labeled data where the correct answers are known.
Unsupervised Learning	Training models on unlabeled data to find hidden structure or patterns.
Feature	A measurable property or characteristic of data used as input for a model (e.g., pixel brightness, word frequency, gene expression level).
Model	A mathematical representation of a system that maps inputs (features) to outputs (predictions).
Loss Function	A formula that quantifies how far the model's predictions are from the true values.
Gradient Descent	An optimization process that updates model parameters to minimize loss by following the slope ("gradient") downhill.
Overfitting / Underfitting	When a model learns noise instead of pattern (overfitting) or fails to capture the structure (underfitting).
Bias-Variance Tradeoff	The balance between model complexity (variance) and generalization (bias).

Activity 1:

What excites and worries you?



1. The Ai possibility that excites me the most in my work is _____

2. The thing about Ai that concerns me the most is _____

https://docs.google.com/document/d/1H0nc5ViBJI-lSij5B_nEZREW70k3P9sCtsV3B1ujb8/edit?usp=sharing

Ai Capabilities & Limitations

What is Ai and how do I work with it?

1. Literacy vs fluency: when to trust, verify, override Ai
2. What Ai is good and bad at
3. What is *learning*?
4. History of Ai

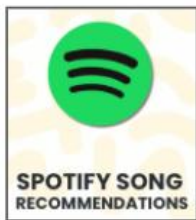
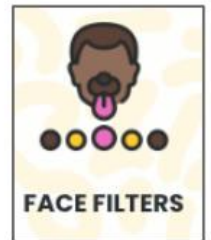
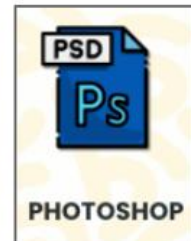
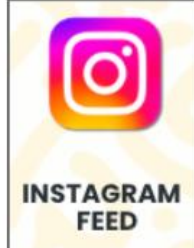
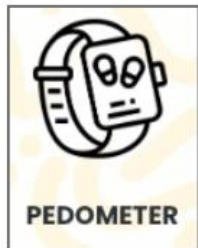


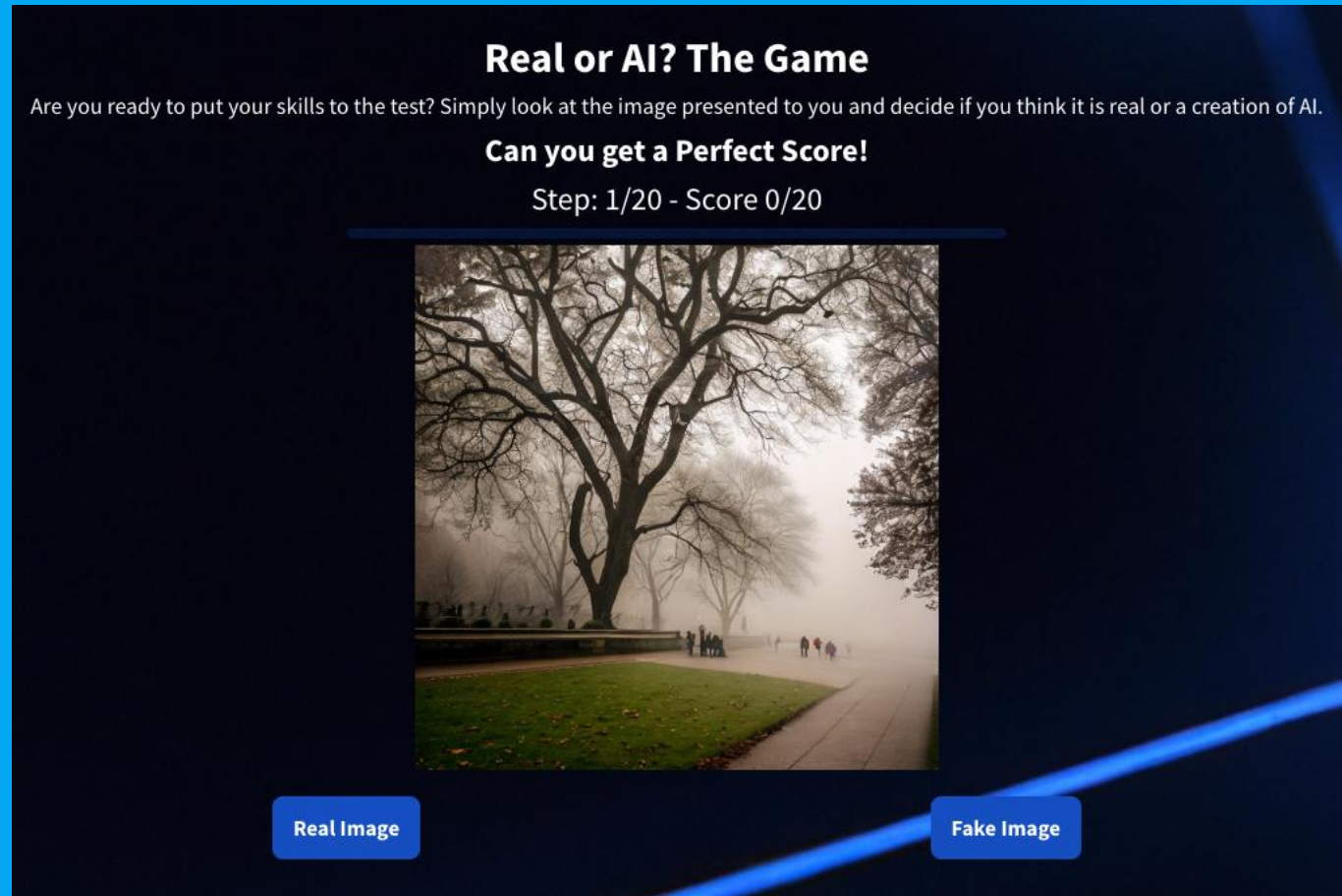
[The Canadian House Hippo](#)

What is Ai?

DEFINITELY
NOT AI

DEFINITELY
AI





Play the “Is it real or is it AI?” Game

What Ai is surprisingly good at

Capability	Example
Pattern Recognition	Detecting tumors, fraud, anomalies
Language Generation	Summaries, drafting, translation
Information Synthesis	Combining many sources into a coherent overview
Prediction & Classification	Risk scores, recommendations, categorization

Ai is good at finding patterns; humans are still better at deciding what those patterns *mean*

What Ai is surprisingly bad at

Hallucination

- Confident but false info
- Fluent $\neq$ factual
- Examples: next slide

Distribution Shift/Reality changes

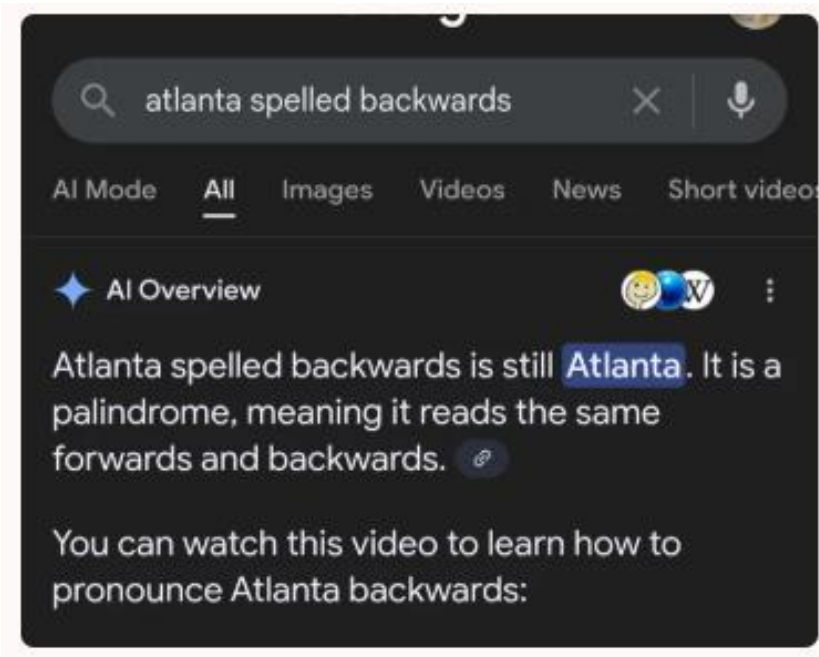
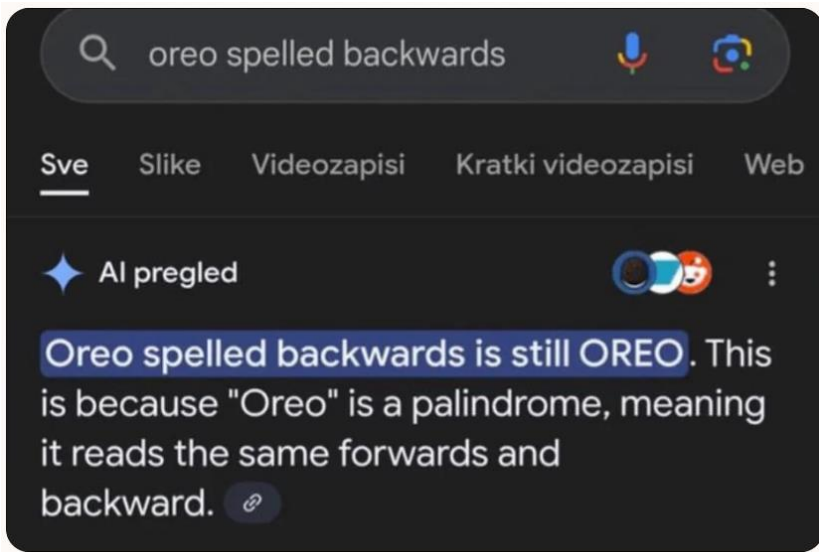
- Ai assumes that tomorrow looks like yesterday (or training data)
- Covid disrupted healthcare models
- New scientific discoveries
- Different populations

Brittleness

- 'intelligence' is often narrower than it appears
- Small changes can cause BIG failures
- Reworded prompt, new context

Optimization Misalignment

- Ai does what you measure, not what you intend
- Engagement algorithms
- Reward hacking



= ChatGPT 5.1 >



Is 73 the 21st prime?

Seventy-three is **not** the twenty-first prime.

Here are the primes around that point:

- 19th prime: **67**
- 20th prime: **71**
- 21st prime: **73**
- 22nd prime: **79**

So 73 is the 21st prime number.

https://www.reddit.com/r/ChatGPT/comments/1pbv54l/73_is_not_the_21st_prime/#lightbox

<https://www.reddit.com/r/mildlyinfuriating/comments/1mnakmj/really/>

What is Ai?

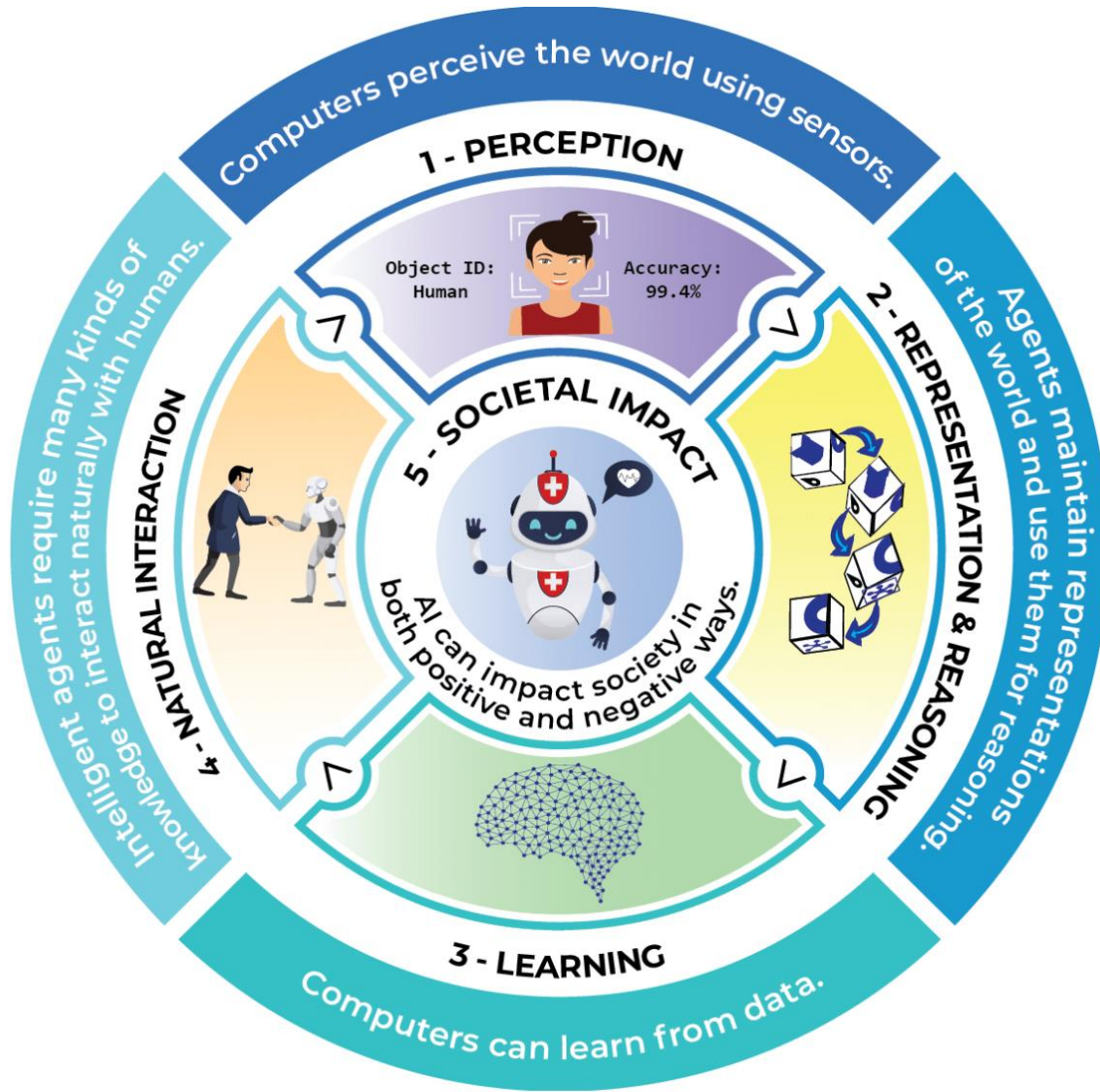
Using Ai versus Understanding Ai

I Can Use AI	I Understand AI
Writes emails	Knows when email content should be verified
Summarizes papers	Understands summaries may omit critical details
Generates code	Recognizes code may contain subtle errors
Answers questions	Knows answers can be hallucinated

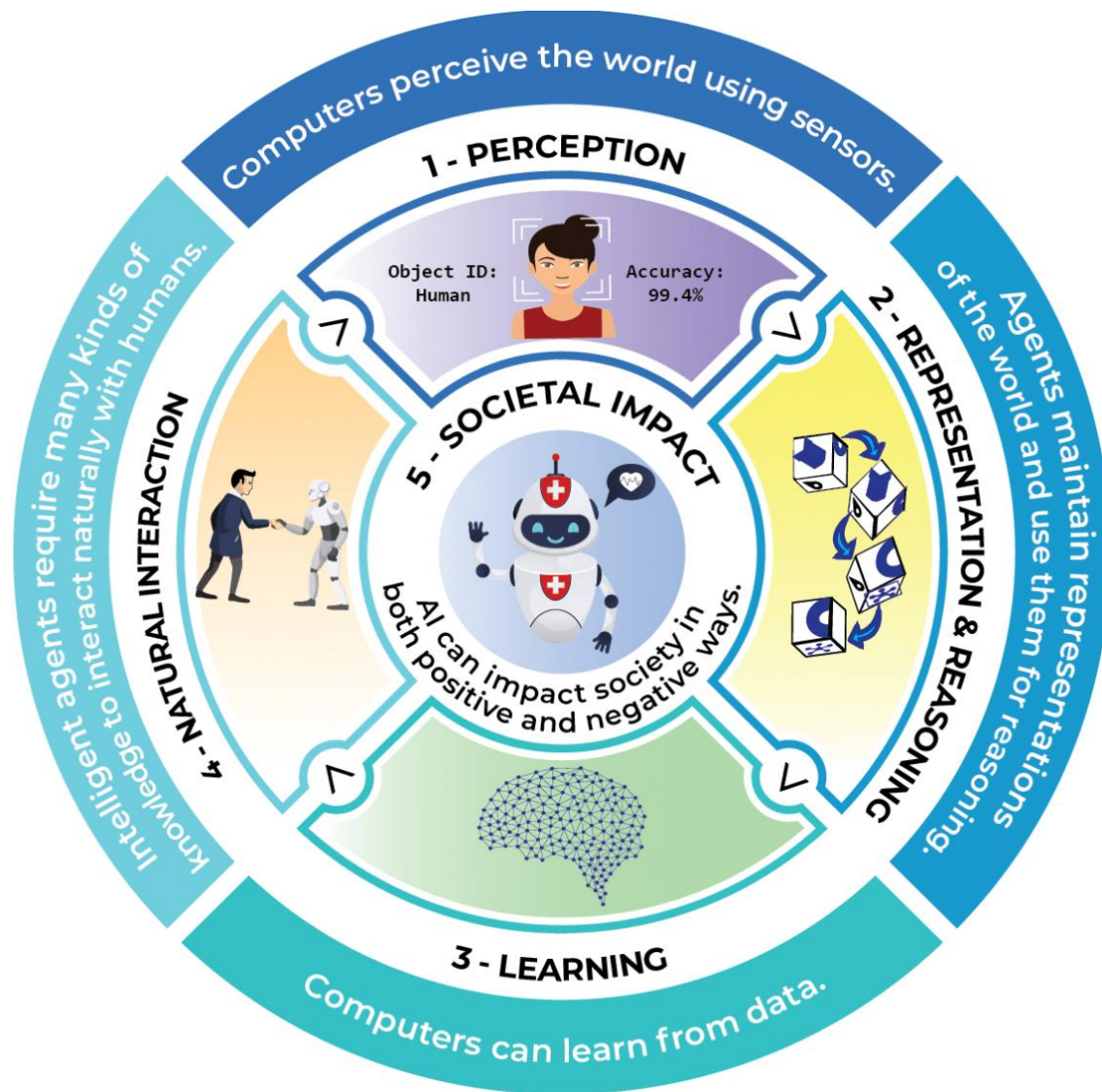
Ai literacy:

knowing when to trust, verify, reject, or override an Ai system





Five Big Ideas Wheel

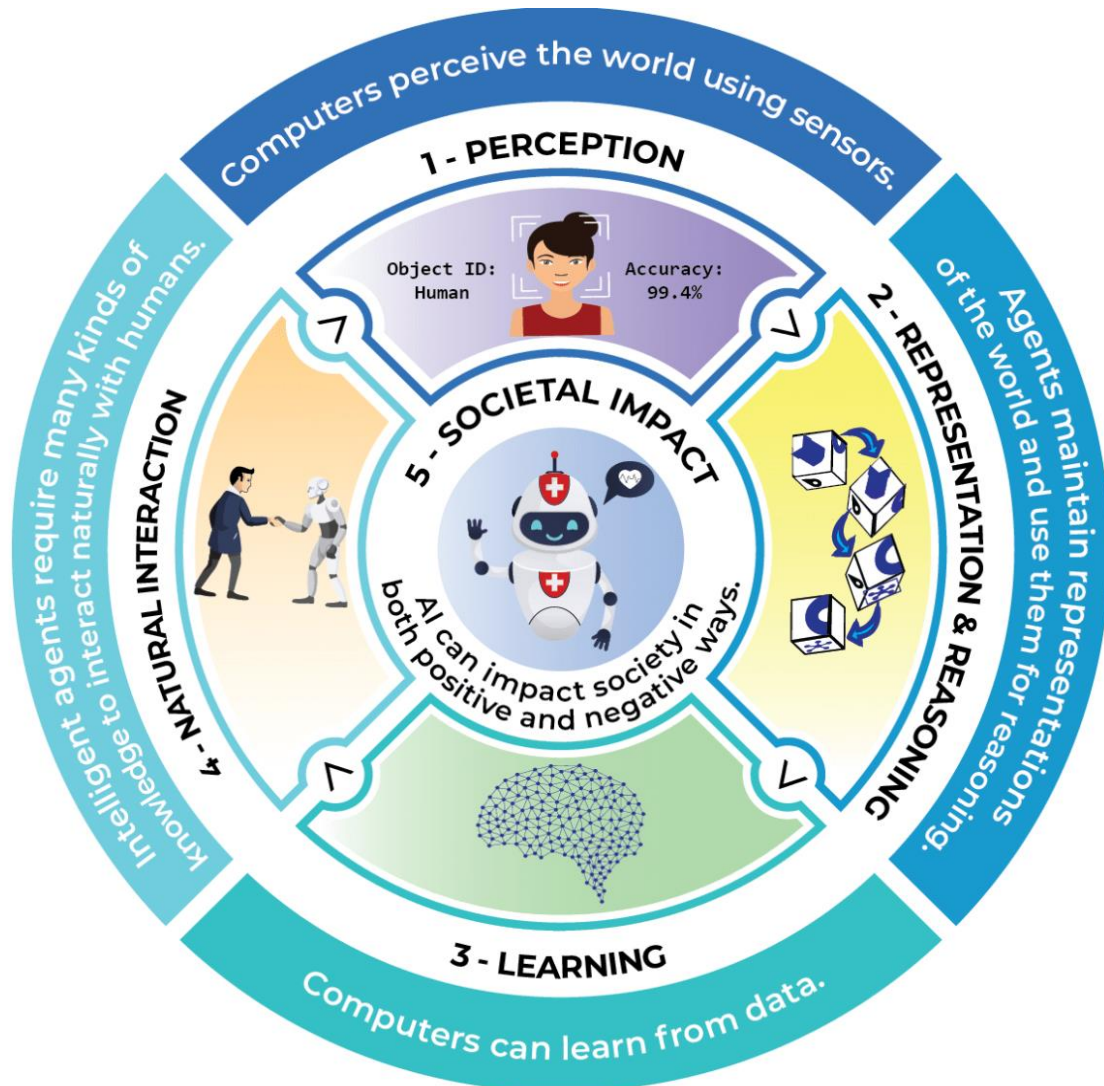


Five Big Ideas Wheel

How do real-world data, like text, images, audio become numerical input?

AI doesn't *eliminate* the need for thinking clearly, it amplifies the consequences of thinking *poorly*.

What *is* learning?



Five Big Ideas Wheel

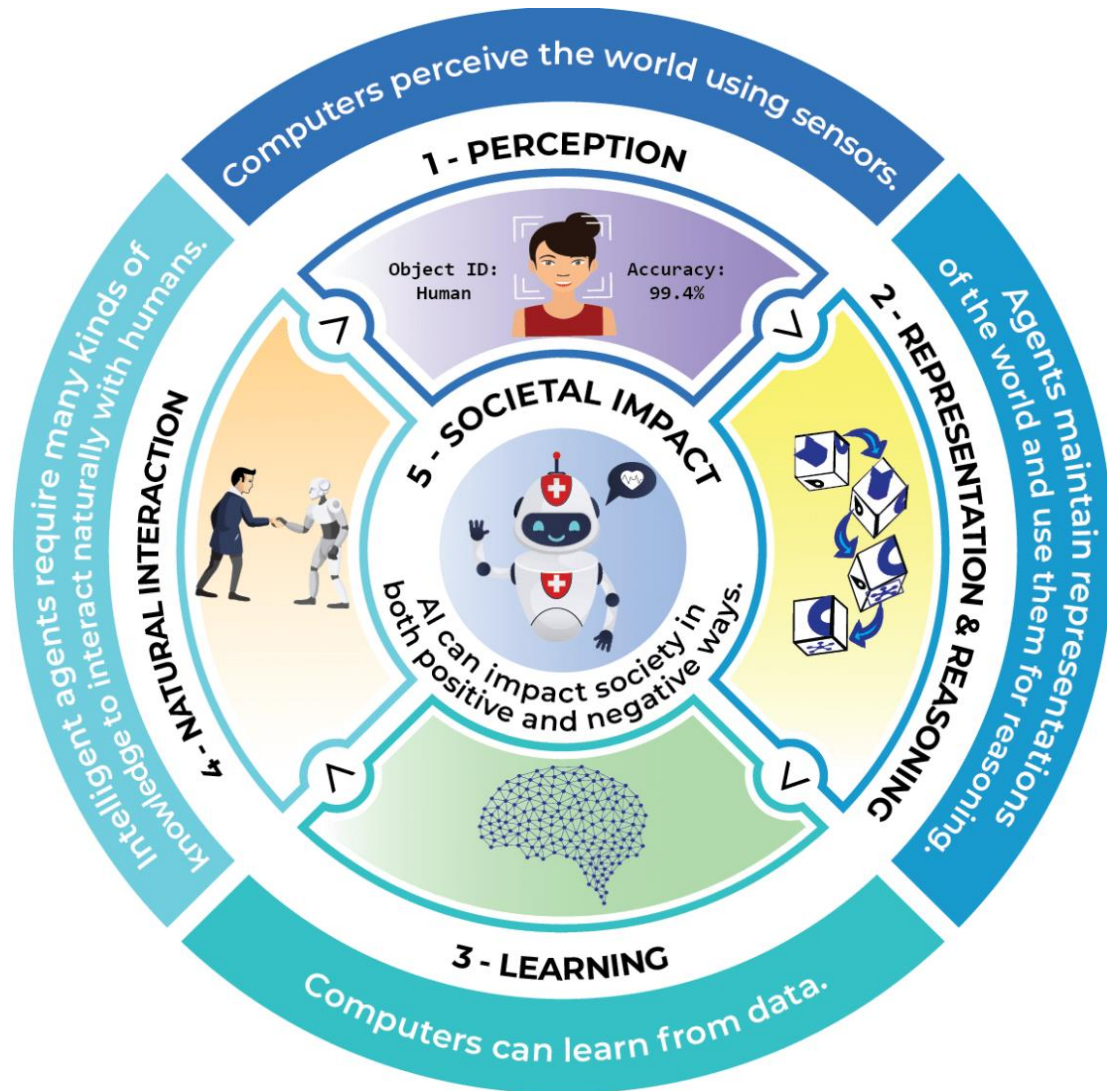
How do real-world data, like text, images, audio become numerical input?



AI doesn't *eliminate* the need for thinking clearly, it amplifies the consequences of thinking *poorly*.



What *is* learning?



How do real-world data, like text, images, audio become numerical input?

- Images → pixel matrices
- Text → tokens → numbers
- Categories → one hot encoding

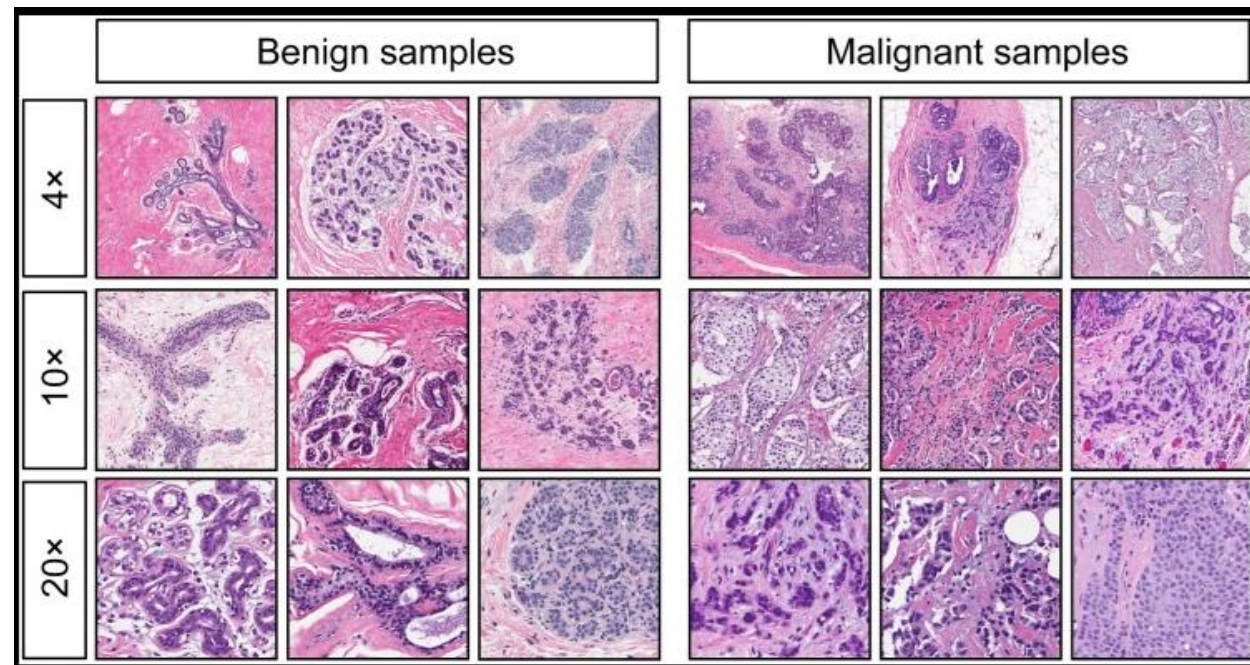
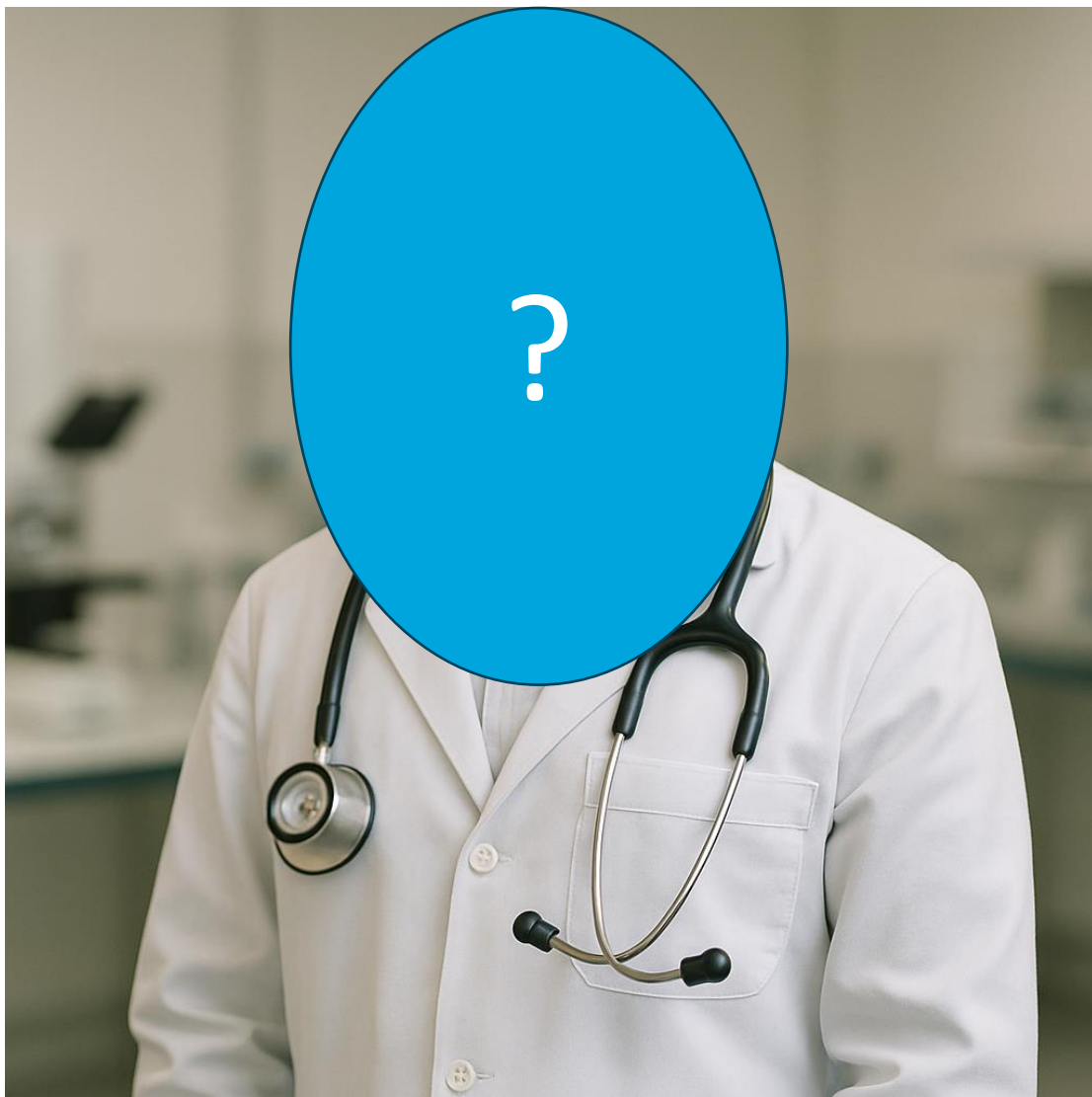


AI doesn't *eliminate* the need for thinking clearly, it amplifies the consequences of thinking *poorly*.



What *is* learning?

Side bar: What *is* learning?



What is learning?



► PLoS One. 2015 Nov 18;10(11):e0141357. doi: [10.1371/journal.pone.0141357](https://doi.org/10.1371/journal.pone.0141357)

Pigeons (*Columba livia*) as Trainable Observers of Pathology and Radiology Breast Cancer Images

[Richard M Levenson](#)^{1,*}, [Elizabeth A Krupinski](#)³, [Victor M Navarro](#)², [Edward A Wasserman](#)^{2,*}

Editor: Jonathan A Coles⁴

► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

PMCID: PMC4651348 PMID: [26581091](https://pubmed.ncbi.nlm.nih.gov/26581091/)

Flock wisdom:

- By day 14, individual birds had 85% accuracy
- A 4 pigeon ‘flock’ had 99% accuracy → consensus

[Home](#) > [Learning & Behavior](#) > [Article](#)

Taking pigeons to heart: Birds proficiently diagnose human cardiac disease

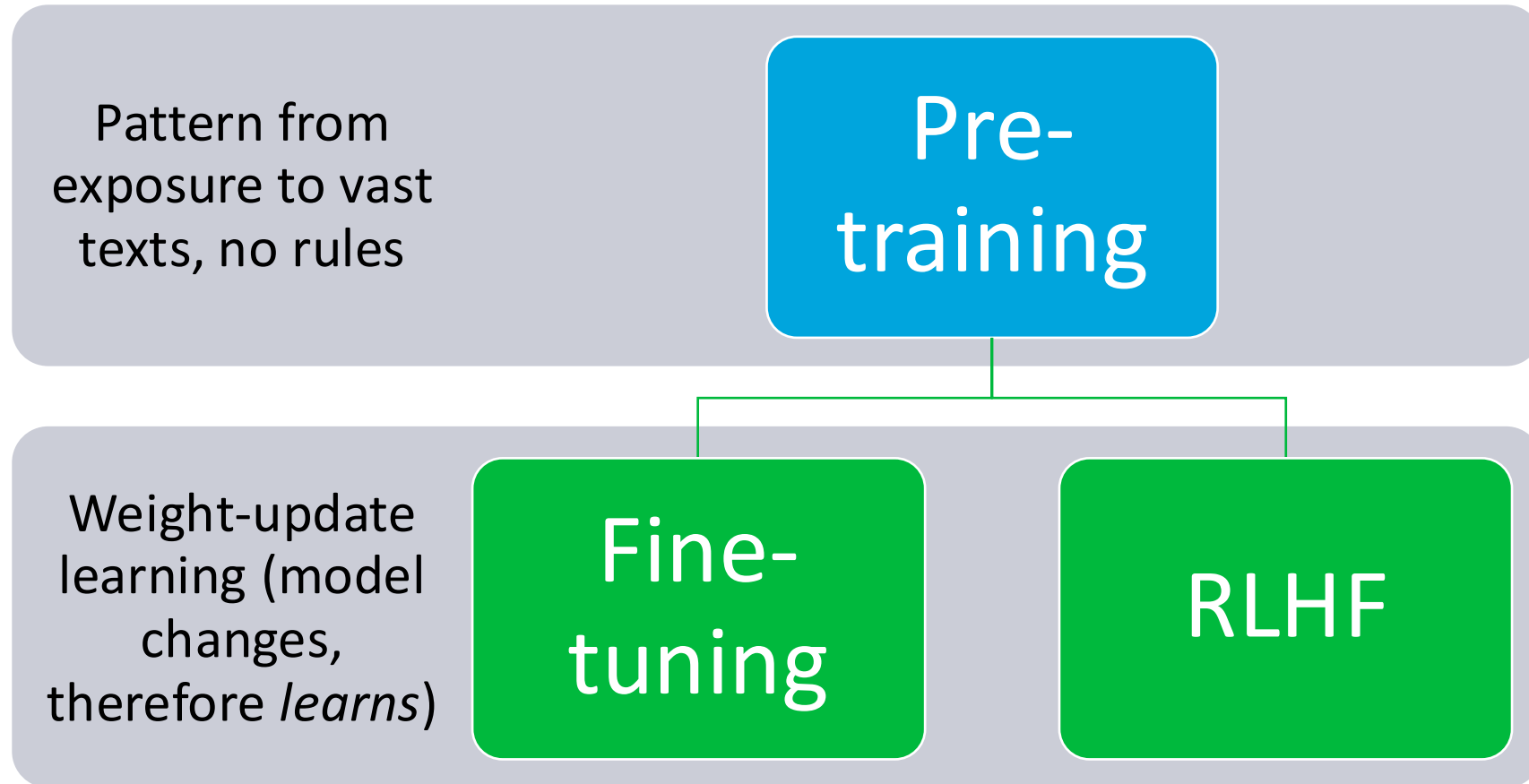
Published: 21 January 2020

Volume 48, pages 9–21, (2020) [Cite this article](#)

* How many “pigeon units” of training does your physician have?

What is learning?

Types of learning in LLMs



Is Artificial Intelligence new?

1943	McCullouch and Pitts - A Logical Calculus of the Ideas Immanent in Nervous Activity - laid foundations for “artificial neurons”
1949	Hebb - “fire together, wire together”
1950	Turing - “Computing Machinery and Intelligence”
1956	McCarthy - Coined the term: Artificial Intelligence - thinking involves both <i>knowledge</i> and <i>reasoning</i> and can be encoded mathematically
1957	Rosenblatt perceptrons → Neural Networks
1967	Minsky
2009	Fei-Fei Li - ImageNet (14 million images labeled)
2012	AlexNet
2024	Nobel Prizes - Physics: Geoffrey Hinton, John J. Hopfield - Chemistry: John J Jumper, David Baker, Demis Hassabis

1308

Catalan poet and theologian [Ramon Llull](#) publishes *Ars generalis* [Artium](#) (The Ultimate General Art), further perfecting his method of using paper-based mechanical means to create new knowledge from combinations of concepts.

1666

Mathematician and philosopher Gottfried Leibniz publishes *Dissertatio de arte combinatoria* (On the Combinatorial Art), following Ramon Llull in proposing an alphabet of human thought and arguing that all ideas are nothing but combinations of a relatively small number of simple concepts.

1726

Jonathan Swift publishes *Gulliver's Travels*, which includes a description of [the Engine](#), a machine on the island of Laputa (and a parody of Llull's ideas): "a Project for improving speculative Knowledge by practical and mechanical Operations." By using this "Contrivance," "the most ignorant Person at a reasonable Charge, and with a little bodily Labour, may write Books in Philosophy, Poetry, Politicks, Law, Mathematicks, and Theology, with the least Assistance from Genius or study."

1763

Thomas Bayes develops a framework for reasoning about the probability of events. [Bayesian inference](#) will become a leading approach in machine learning.

1854

[George Boole](#) argues that logical reasoning could be performed systematically in the same manner as solving a system of equations.

1898

At an electrical exhibition in the recently completed Madison Square Garden, Nikola Tesla makes a demonstration of [the world's first radio-controlled vessel](#). The boat was equipped with, as Tesla described, "a borrowed mind."

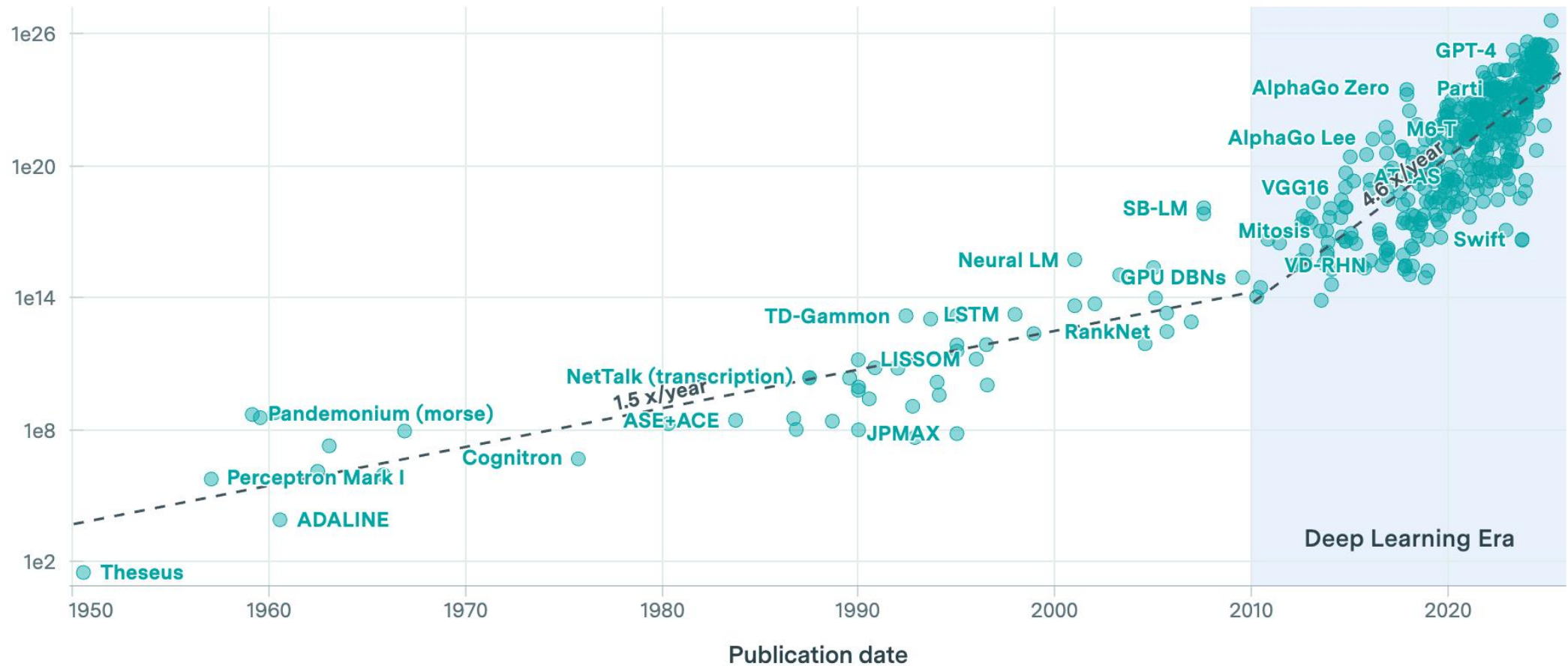
1914

The Spanish engineer [Leonardo Torres y Quevedo](#) demonstrates the first chess-playing machine, capable of king and rook against king endgames without any human intervention.

Notable AI Models

Training compute (FLOP)

469 Results



CC-BY

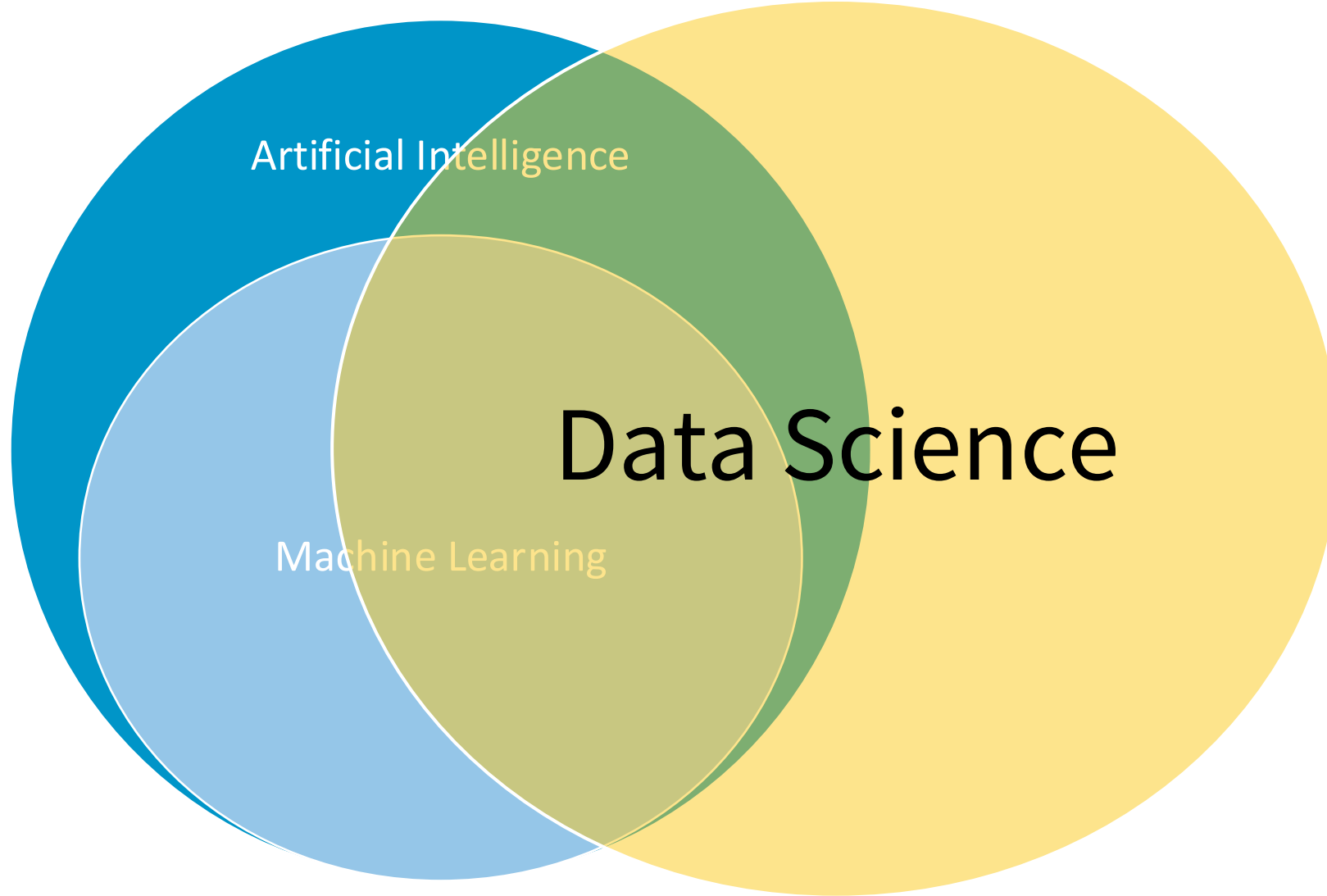
epoch.ai

15-minute Break

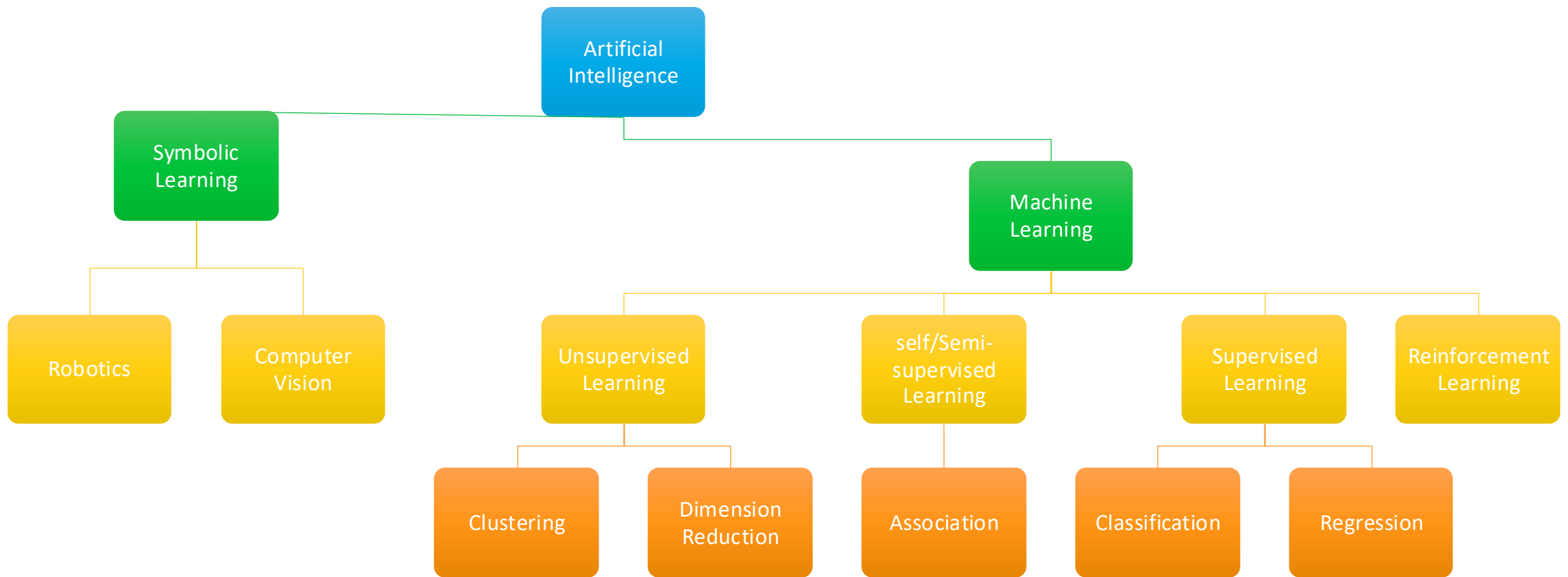
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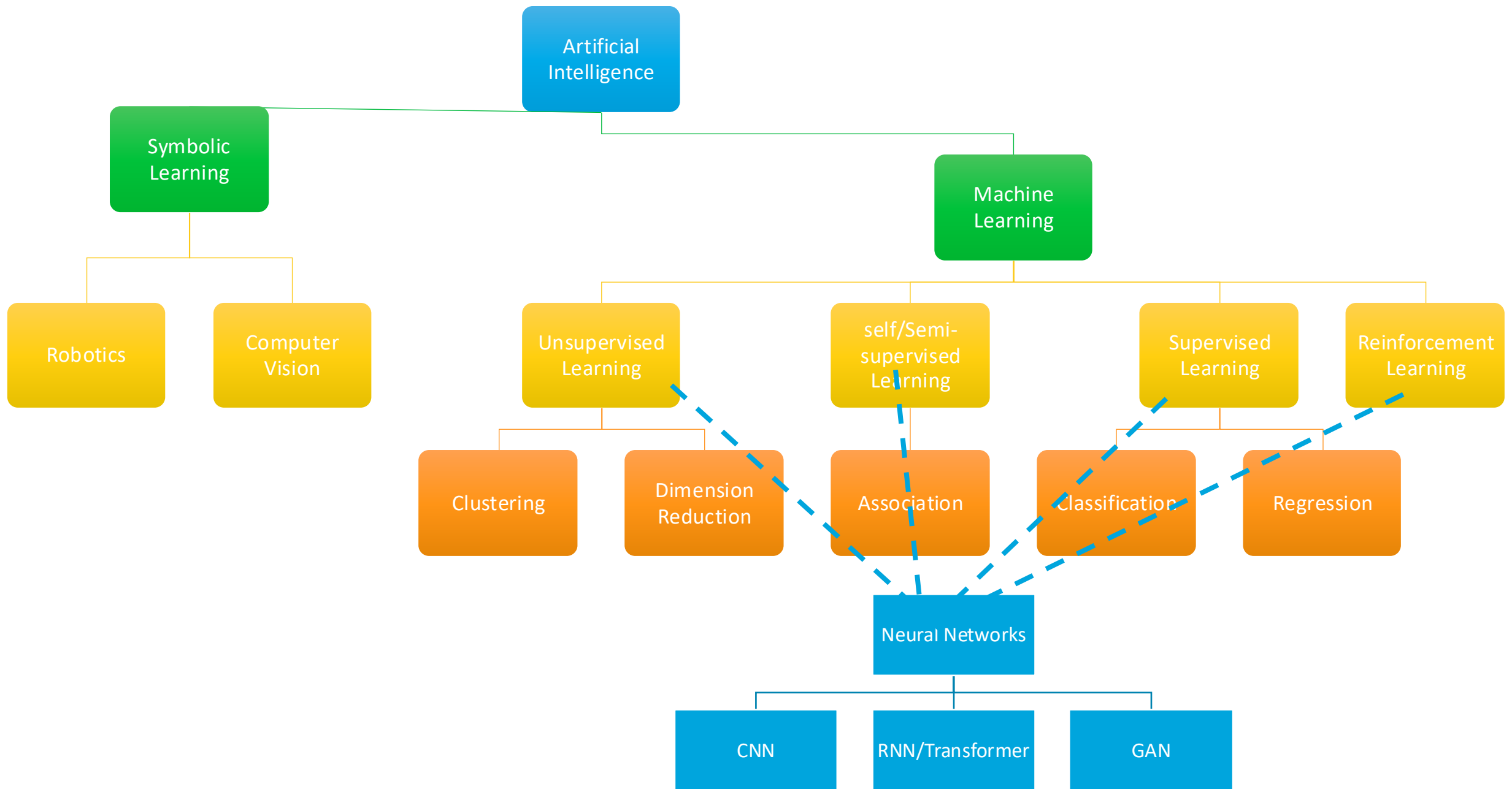
How are Ai, Machine Learning, and Data Science related?

- Definitions
- Breakdown of Ai into ML categories



Data Science = Statistics + Linear Algebra + Computer Science + Domain Data

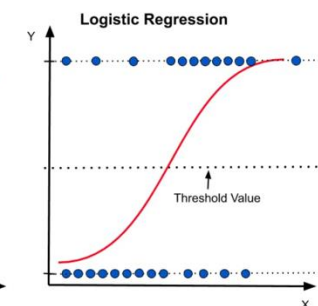
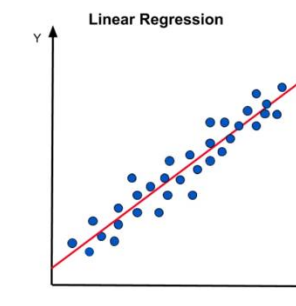
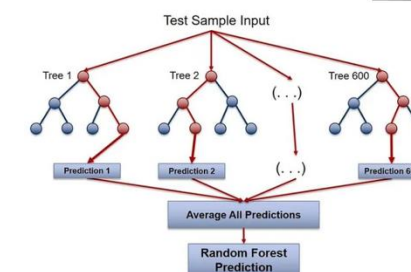
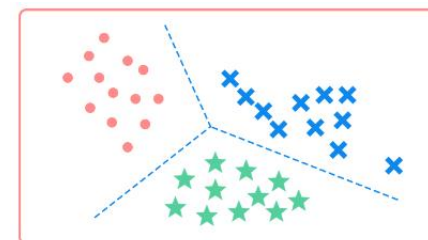
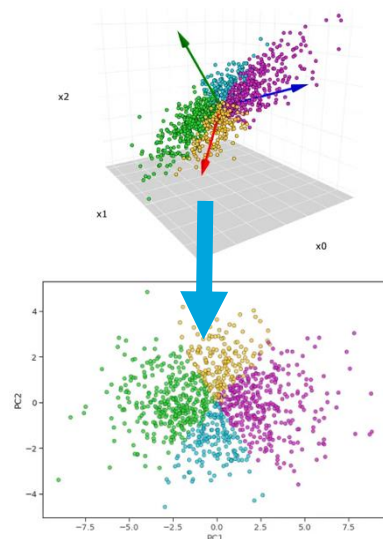
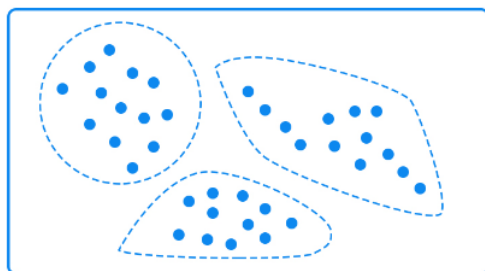
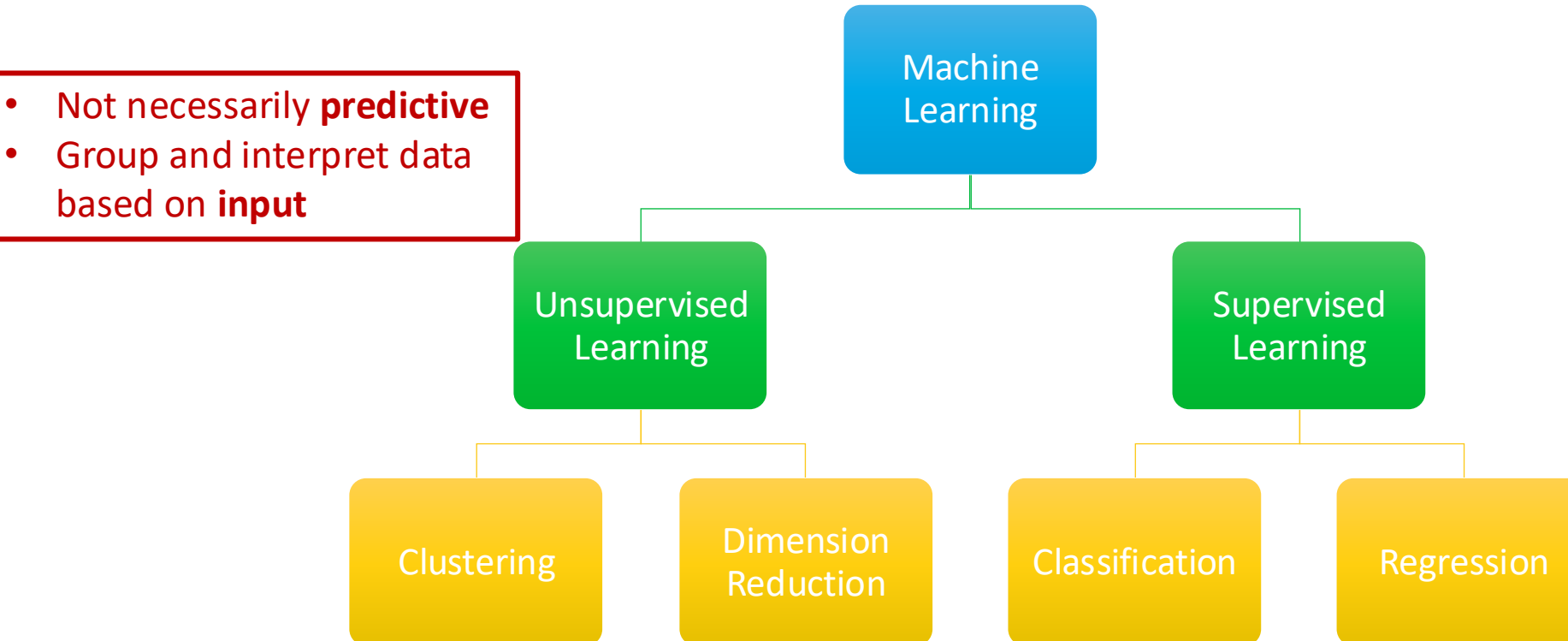




- Not necessarily **predictive**
- Group and interpret data based on **input**

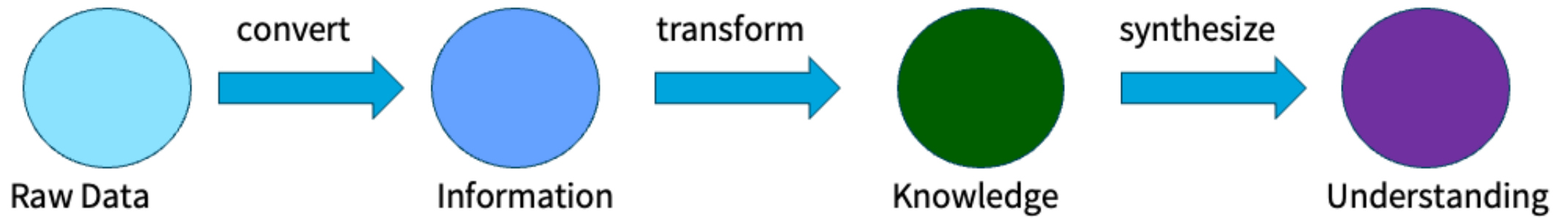
Develop **predictive model** based on **input** and **output**

Data is labelled

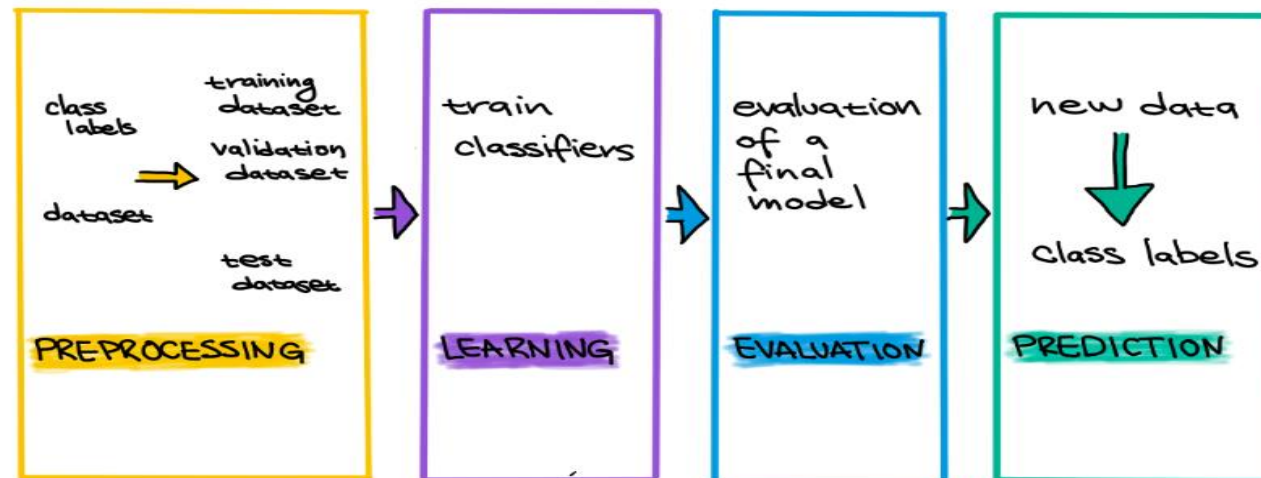


The Features Game

How do we translate the world into
computationally tractable features?



Machine learning workflow



Raw Data Becomes Numbers:

- All inputs – pixels, words, sounds – must be converted into numbers so that models can process them

Images

Pixels → RGB values → Matrices

Categories

Labels → One hot Encoding

Text

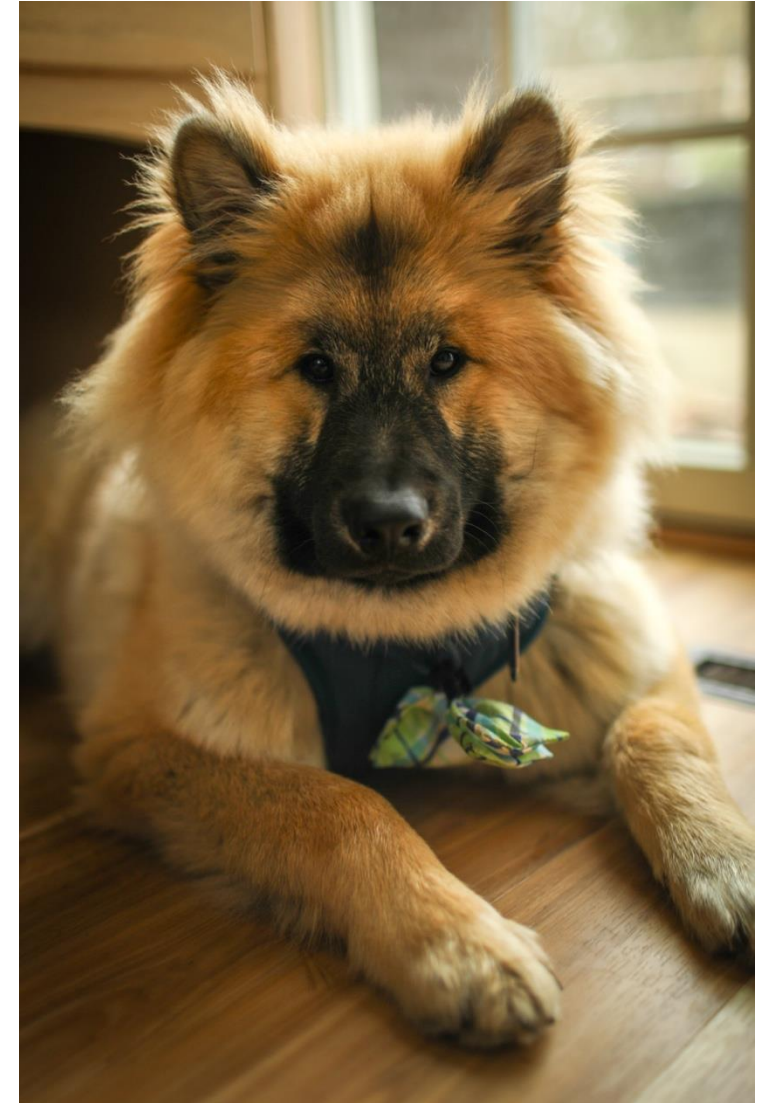
Words → Tokens → Embeddings

Audio

Waveforms → Frequencies → Vectors

How does raw data become numbers?

Huxley is an 18-month-old Eurasier (male, not neutered) who came in on 11/03/2025 for a sick visit because his owner said he hasn't been eating normally and vomited twice. The tech wrote his weight as **40 pounds**, though later in the notes someone jotted “~18 kg?” with a question mark next to it. On exam, Huxley's temperature was 102.1 F, heart rate 96 bpm, and his respiratory rate was listed as “**about 30/min**,” although the structured intake sheet earlier said 28/min. His body condition score was 6/9 and hydration was estimated around 5% dehydrated. Pain was mild (1 on the 0–4 scale). CBC showed WBC 14.2 (slightly high), RBC 6.3, hemoglobin 15.1 g/dL, and platelets 268. Chemistry panel had BUN 21, creatinine 1.1, ALT 68 (a bit high), and glucose 87. Abdominal radiographs noted some gastric fluid but no foreign body. The assessment was possible gastritis with mild dehydration, no obstruction suspected. Plan: maropitant at 1 mg/kg once daily for 3 days, 200 mL SQ fluids, bland diet for 2–3 days with small frequent meals and return if vomiting continues for more than 48 hours.

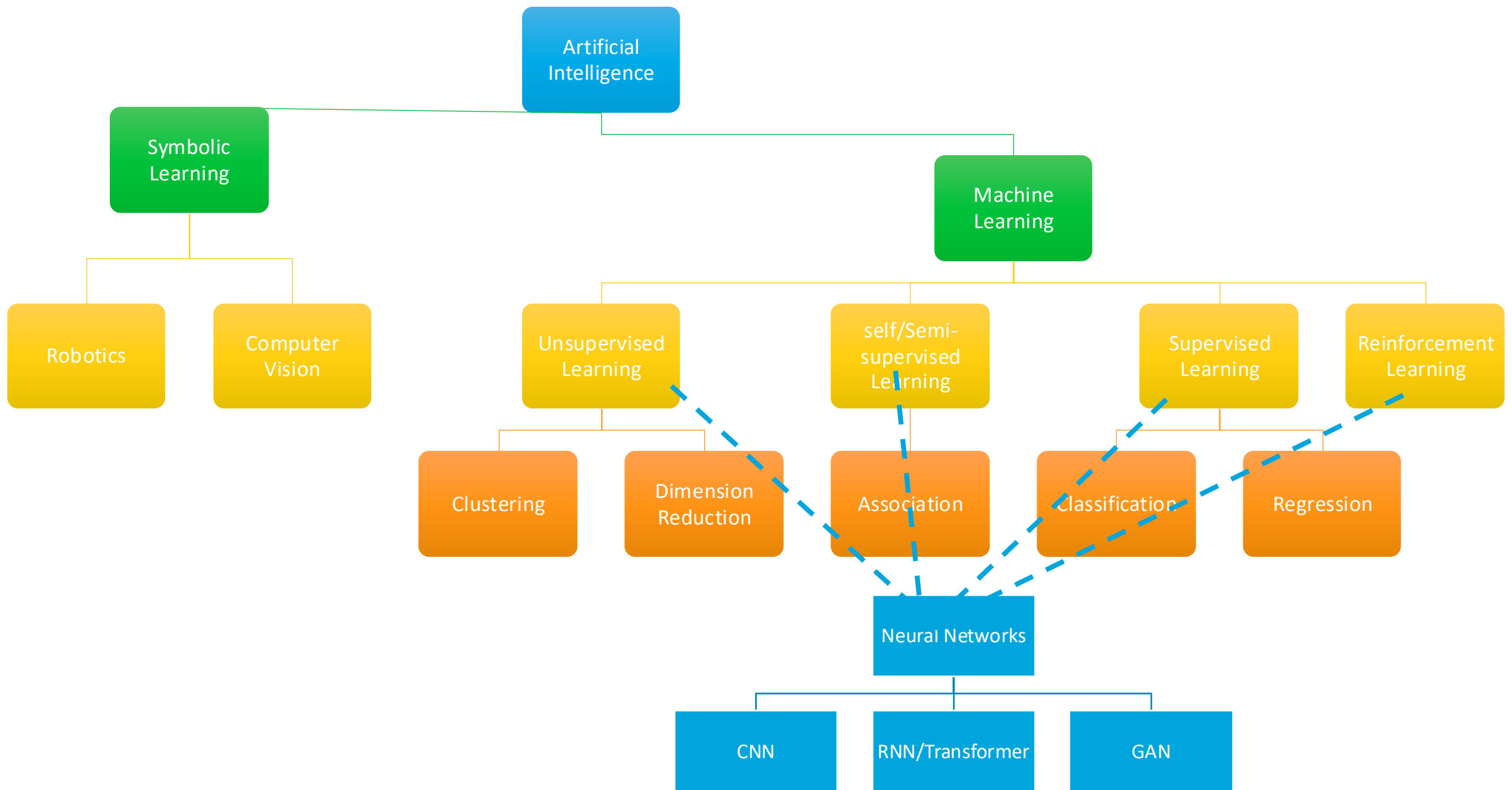


15-minute Break

Return at:

How Machines learn: Regression, k-means & loss function

- Building intuition for the machinery of everything that we will discuss this week



Machine Learning

Unsupervised Learning

Supervised Learning

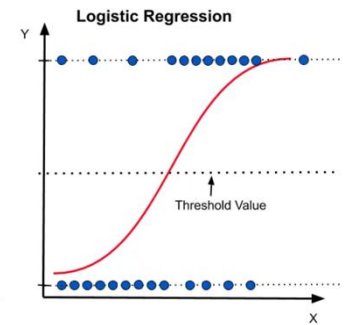
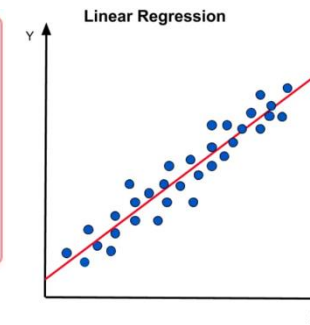
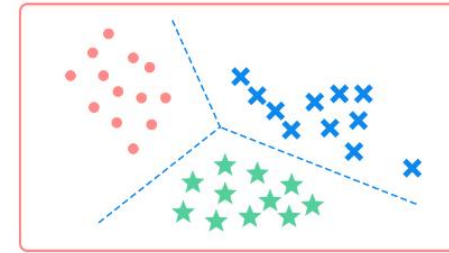
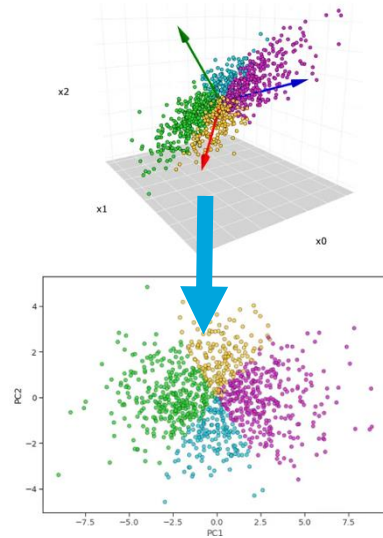
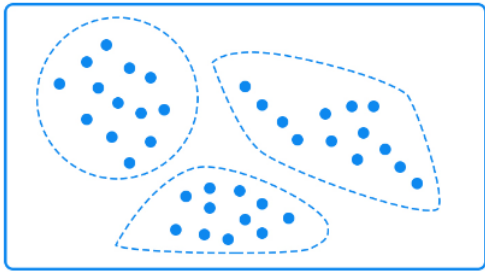
Clustering

Dimension Reduction

Association

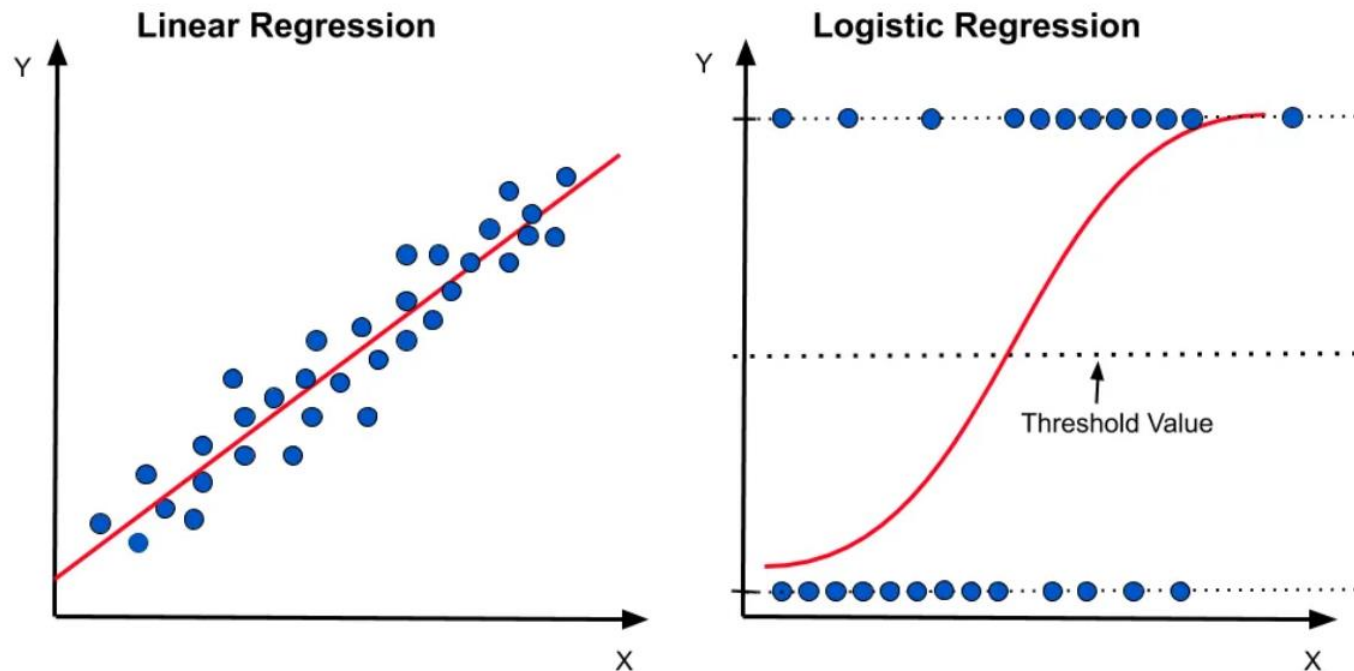
Classification

Regression

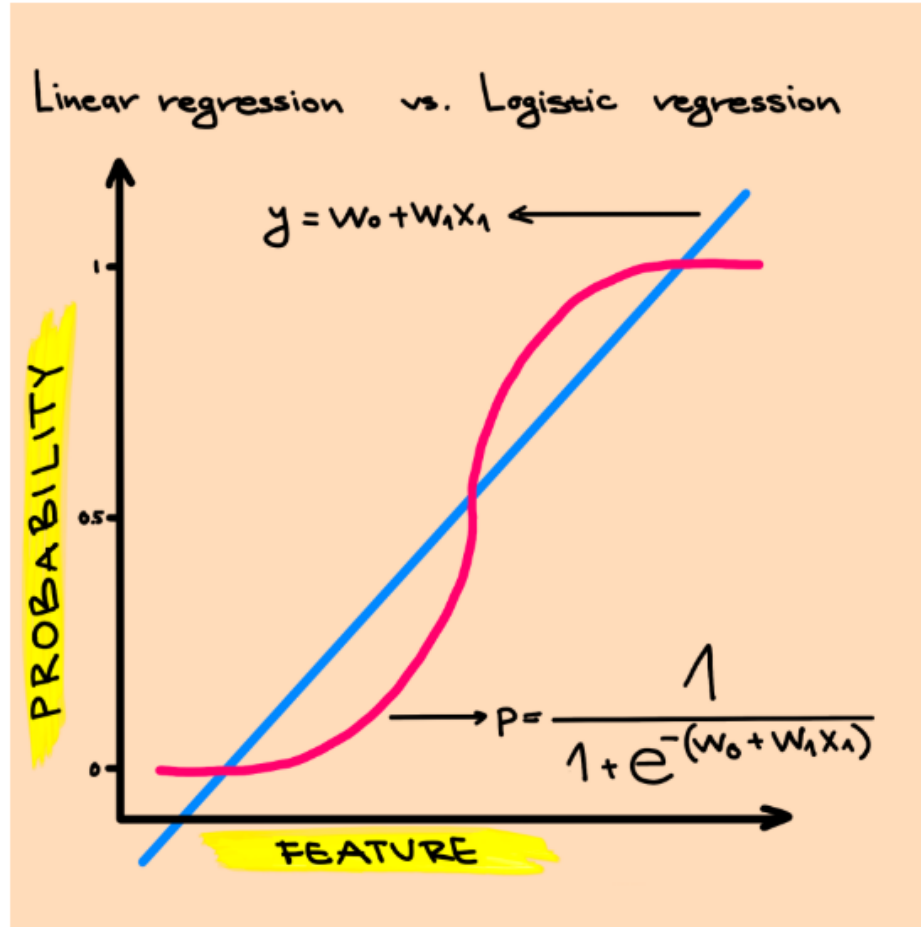


In machine learning, computers learn patterns that help them to make decisions without being explicitly programmed.

- A familiar way to make predictions is with regression:



Supervised ML: **Logistic & Linear Regression:**



<https://carpentries-incubator.github.io/ml4bio-workshop/05-logit-ann/index.html>

Questions to consider for Logistic Regression:

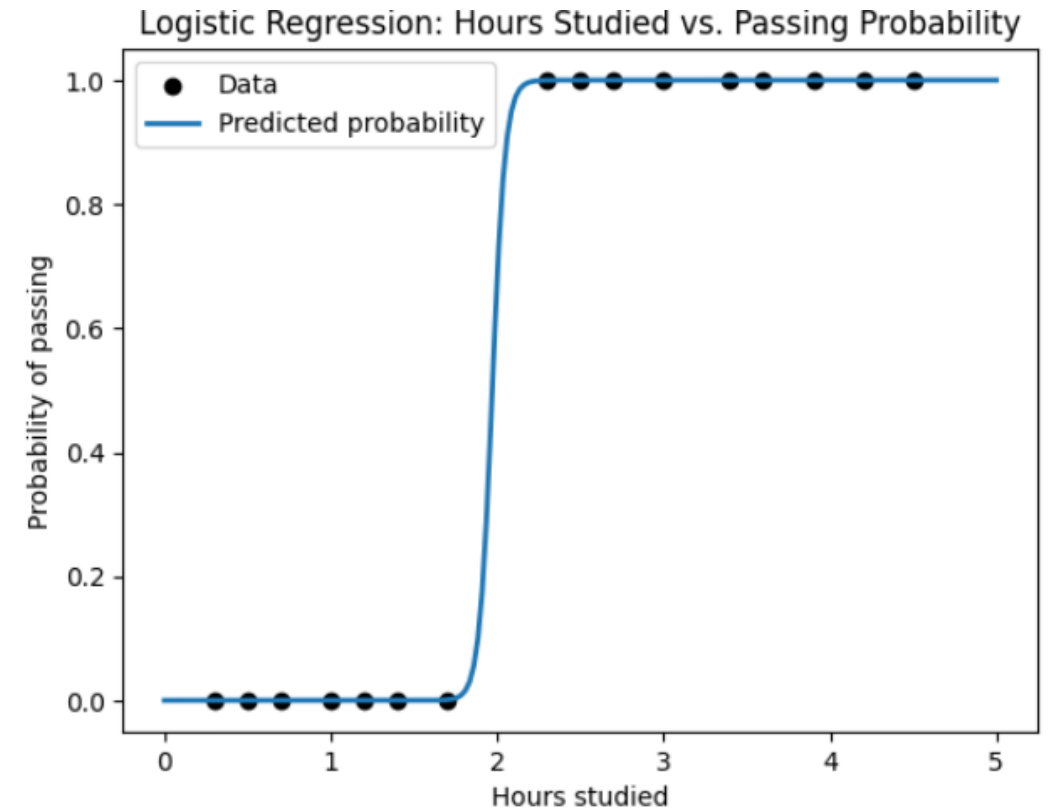
- How do Ai systems use features (input variables) to generate predictions?
- How do supervised models make decisions?
- Can we impact the decision that it makes?
- Consider where logistic regression models are used (healthcare, criminal justice, finance). What are the risks of defining the cut-off value? Of including certain features and not others?

Supervised ML: **Logistic Regression:**

Hours studied	Result
0.3, 0.5, 0.7, 1.0, 1.2, 1.4, 1.7	Fail
2.3, 2.5, 2.7, 3.0, 3.4, 3.6, 3.9, 4.2, 4.5	Pass

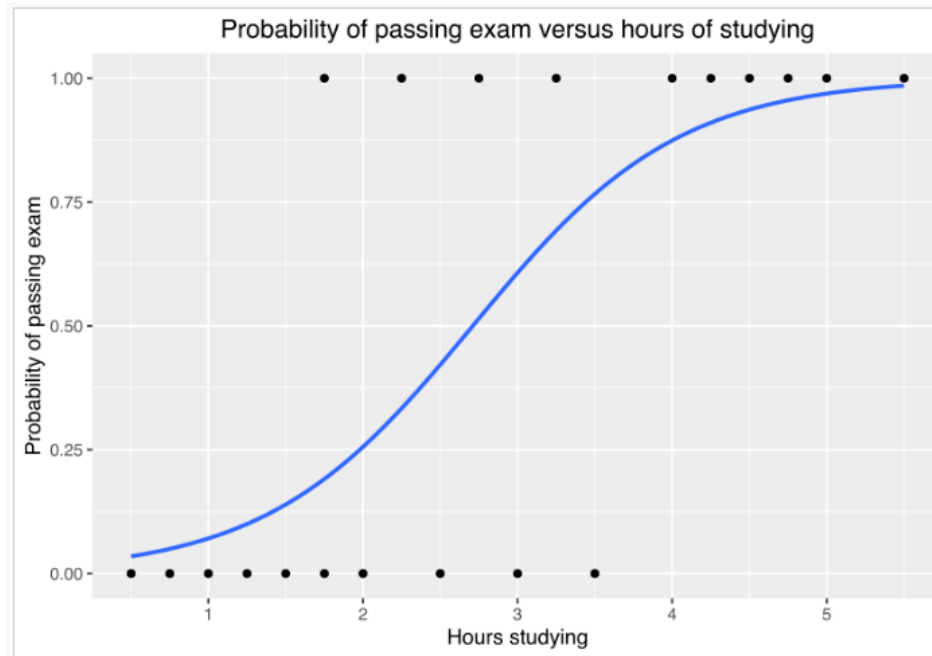
Supervised ML: **Logistic Regression:**

Hours studied	Result
0.3, 0.5, 0.7, 1.0, 1.2, 1.4, 1.7	Fail
2.3, 2.5, 2.7, 3.0, 3.4, 3.6, 3.9, 4.2, 4.5	Pass

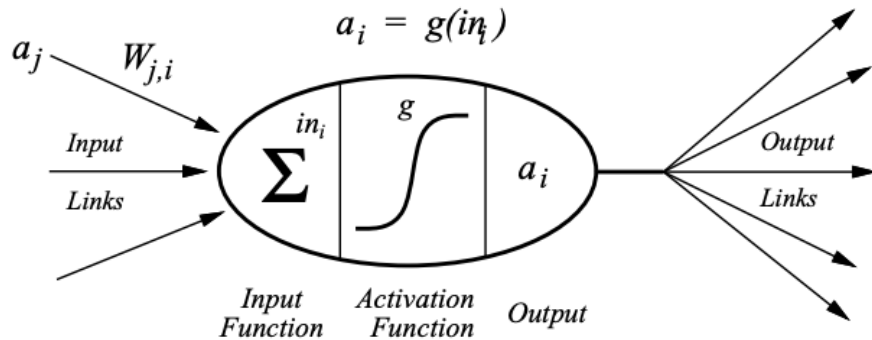
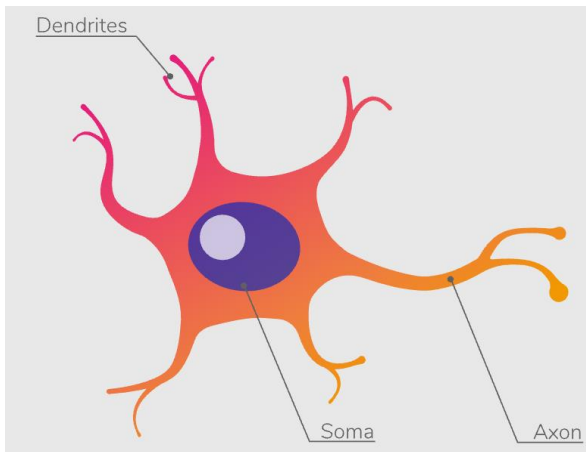


Supervised ML: **Logistic Regression:**

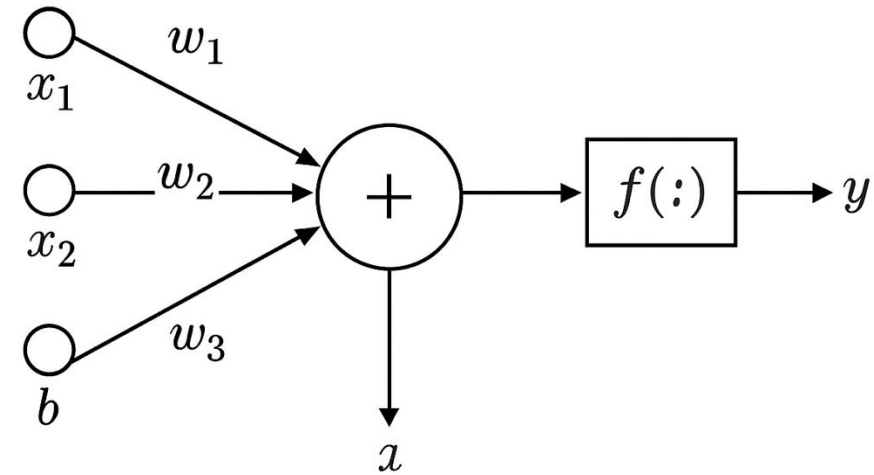
Hours studied	Result
0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0, 2.5, 3.0, 3.5	Fail
1.75, 2.25, 2.75, 3.25, 4.0, 4.25, 4.5, 4.75, 5.0, 5.5	Pass



Graph of a logistic regression curve fitted to the (x_m, y_m) data. The curve shows the probability of passing an exam versus hours studying.



$$a_i = g\left(\sum_j W_{j,i} a_j\right)$$



Inputs+ Bias + weights → hidden layers → predictions

Question to consider for The Weight Adjustment Game:

- How do we get the optimal weights?

Question to consider for Fair Treasure Splitting Game:

- What trade-offs were you willing to accept?

How does a model know it's wrong? How do we measure 'wrongness'?

The Weight Adjustment Game

Age	Glucose	Has Diabetes (Yes = 1, No = 0)
40	130	1
25	100	0

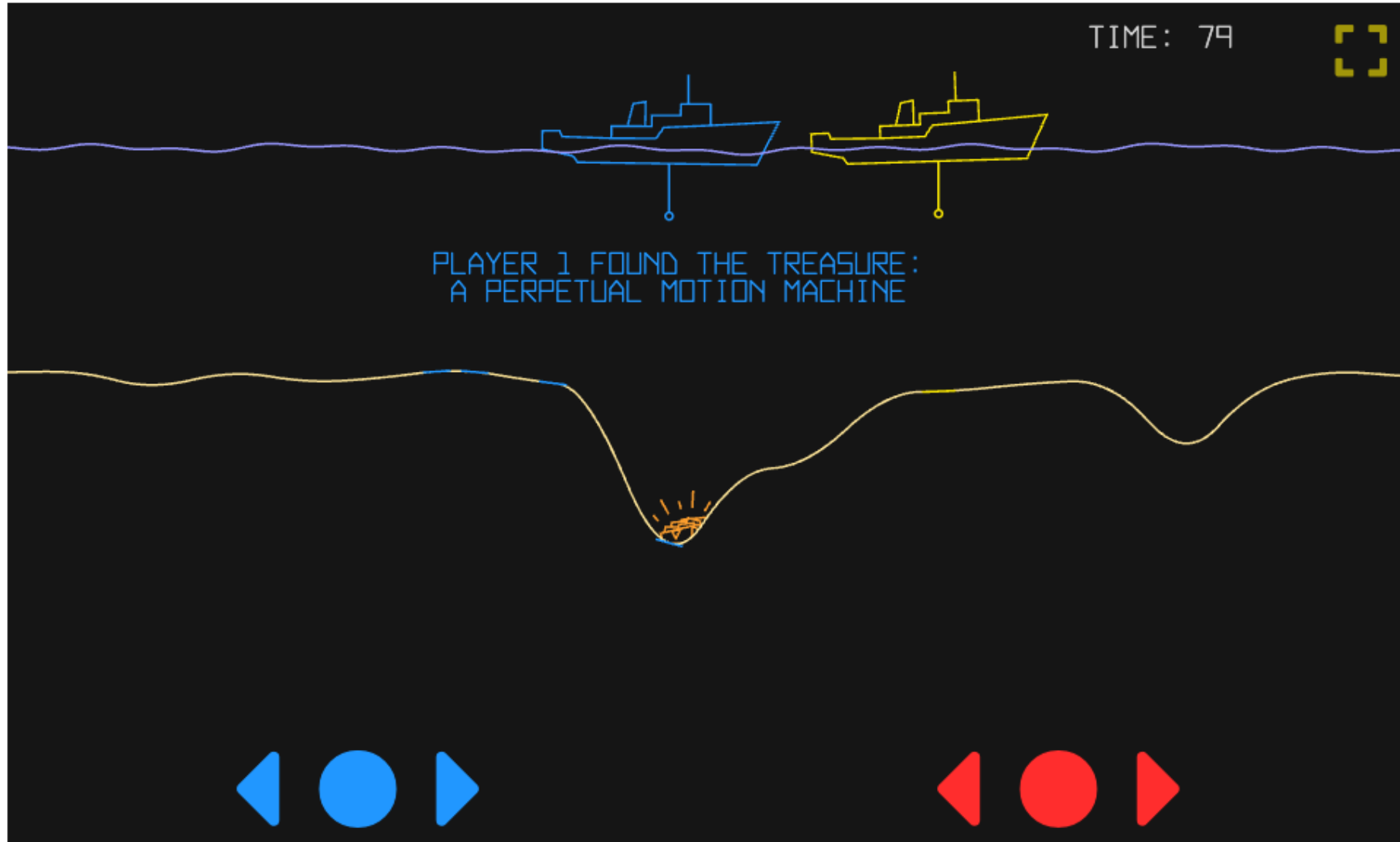
Fair Treasure Splitting Game

1. Your group will receive one set of diamonds
2. You need to split the diamonds into 2 groups
3. Calculate the total value of each pile
4. Determine the **loss** (difference in values)
5. The team with the **smallest loss** wins

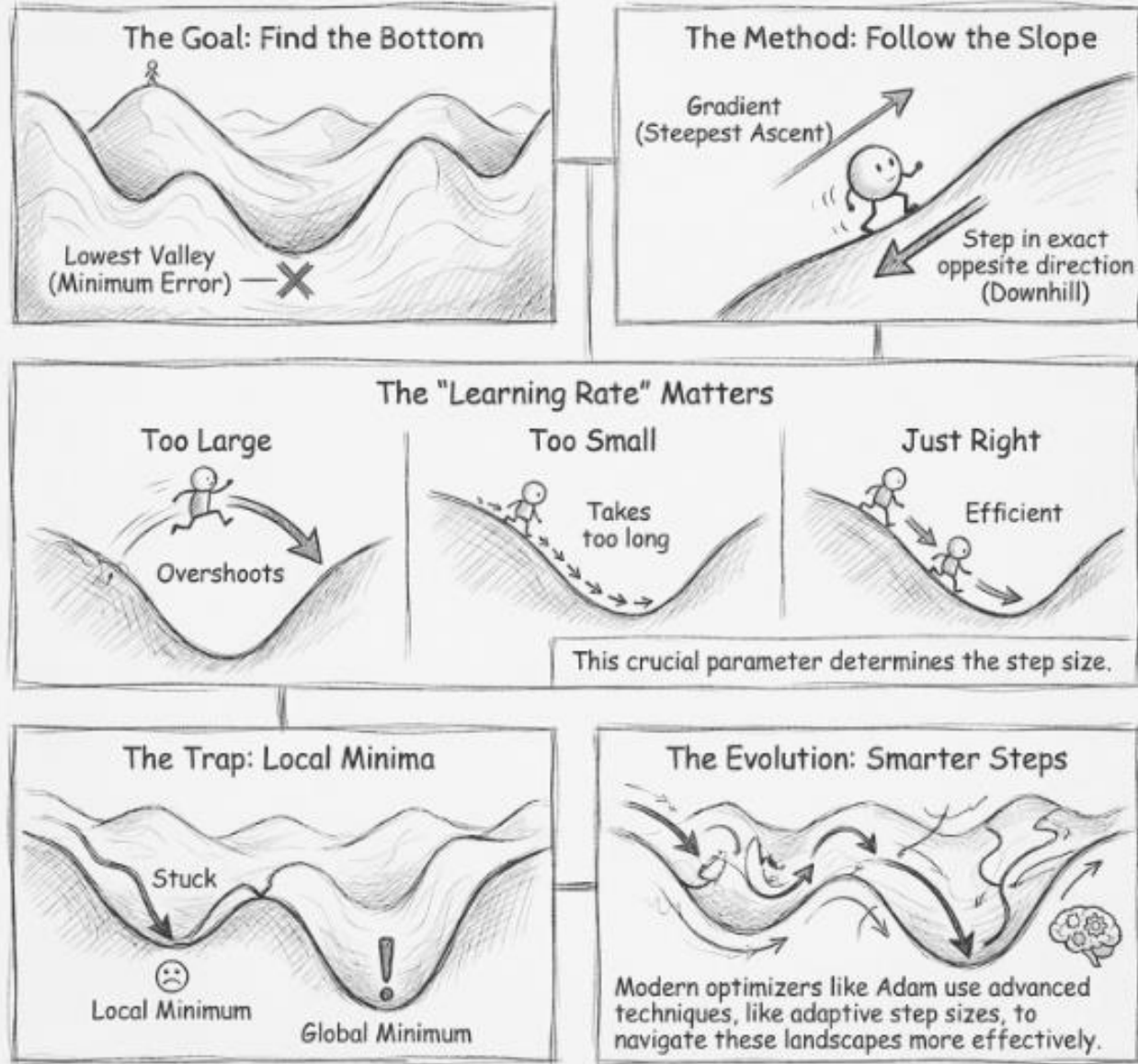
Thing to think about:

- What was your strategy for dividing the diamonds?
- Was your split fair?
- If given more time, is there another approach that you might take?

Gradient Descent



How Machines Learn: The Art of Gradient Descent



You an Ai system. Your only job is to predict a number.

Pick a number between 0-100.



You are an AI system. Your only job is to predict a number. Pick a number between 0-100.

Now I will tell you the loss function.

$$\text{Loss} = |\text{prediction} - 50|$$

What are people's choices?

You an Ai system. Your only job is to predict a number.

Pick a number between 0-100.

What if the loss is now calculated this way:

$$\text{Loss} = (\text{prediction} - 50)^2$$

Would you change your original choice?

You an Ai system. Your only job is to predict a number.

Pick a number between 0-100.

What if the loss is now calculated this way:

Loss:

- **If prediction $> 50 \rightarrow$ penalty x 5**
- **if prediction $< 50 \rightarrow$ normal**

Which number would you pick now and why?

Debrief:

- You didn't become a different Ai system between rounds
- There was no new information given
- The only change is what you were 'punished' for

Questions to consider for the following simulation:

- Why are k-means clustering *unsupervised*?
- Can unsupervised algorithms uncover real patterns?
- How does k-means demonstrate how learning is done via algorithms?
- Are these clusters ‘real’?
- Parameter ‘knobs’:
 - Does it matter how many k are chosen (k=2,k=3,etc)
 - Do different initial placements change the results?

Unsupervised ML: **Classification**

Visualizing K-Means Clustering

January 19, 2014

Suppose you plotted the screen width and height of all the devices accessing this website. You'd probably find that the points form three clumps: one clump with small dimensions, (smartphones), one with moderate dimensions, (tablets), and one with large dimensions, (laptops and desktops). Getting an algorithm to recognize these clumps of points without help is called *clustering*. To gain insight into how common clustering techniques work (and don't work), I've been making some visualizations that illustrate three fundamentally different approaches. This post, the first in this series of three, covers the k-means algorithm. To begin, click an initialization strategy below:

How to pick the initial centroids?

I'll Choose

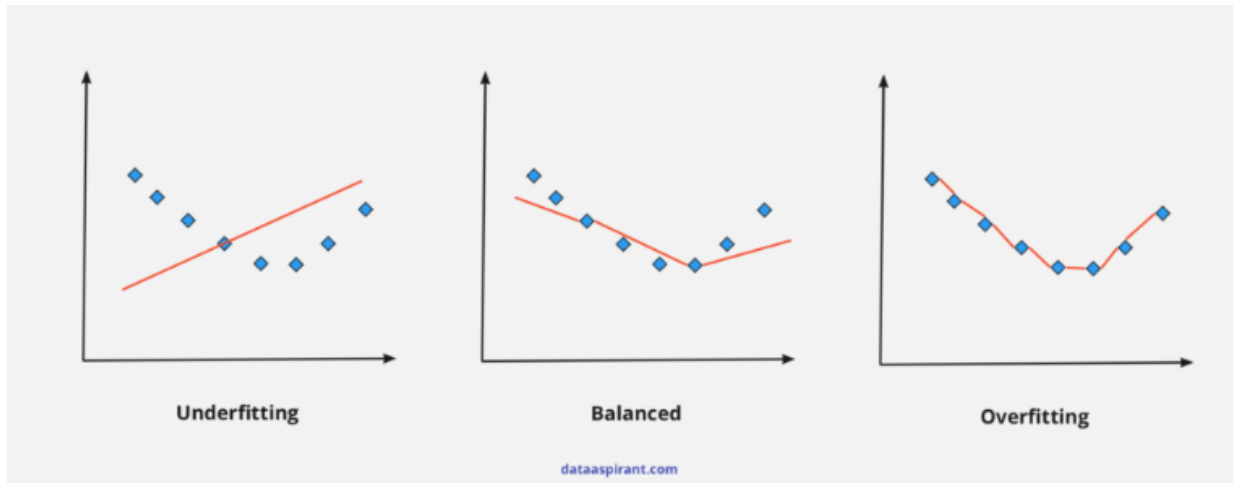
Randomly

Farthest Point

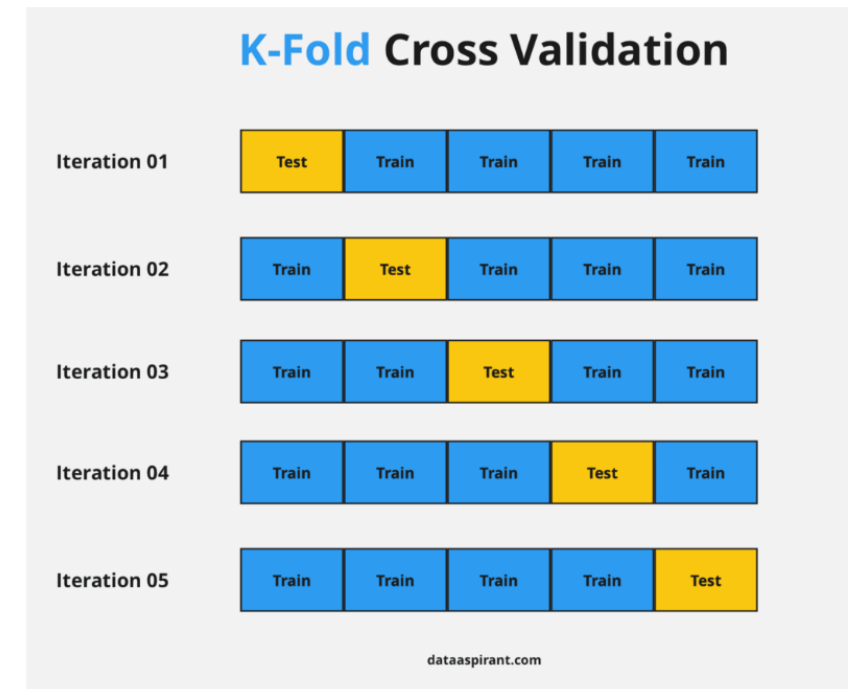
Training versus Testing

- Strategy:

split the data and use some of it for **training** the NN to **make predictions** and **reserve** some of it **to test** the NN



- K-fold cross-validation



Hours Studied	Result
0.5	Fail
0.75	Fail
1.0	Fail
1.25	Fail
1.5	Fail
1.75	Fail
1.75	Pass
2.0	Fail
2.25	Pass
2.5	Fail
2.75	Pass
3.0	Fail
3.25	Pass
3.5	Fail

Hours Studied	Result
4.0	Pass
4.25	Pass
4.5	Pass
4.75	Pass
5.0	Pass
5.5	Pass

<u>Training</u>	<u>Testing</u>
1,2,3,4	5
1,2,3,5	4
1,2,4,5	3
1,3,4,5	2
2,3,4,5	1

K=5 → **Fold 1:** 0.5, 2.25, 3.25, 4.75
Fold 2: 0.75, 1.75, 2.75, 4.5
Fold 3: 1.0, 2.5, 3.0, 4.25
Fold 4: 1.5, 1.75, 4.0, 5.5
Fold 5: 1.25, 2.0, 3.5, 5.0

Synthesis of the day's material

1. Ai isn't 'new'; it is mostly Machine Learning
2. Stacked logistic regressions approximates neural networks

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1. Ai isn't 'new'; it is mostly ML
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3. Ai systems = Data x

(Representation + Loss) x Optimization

How data is encoded

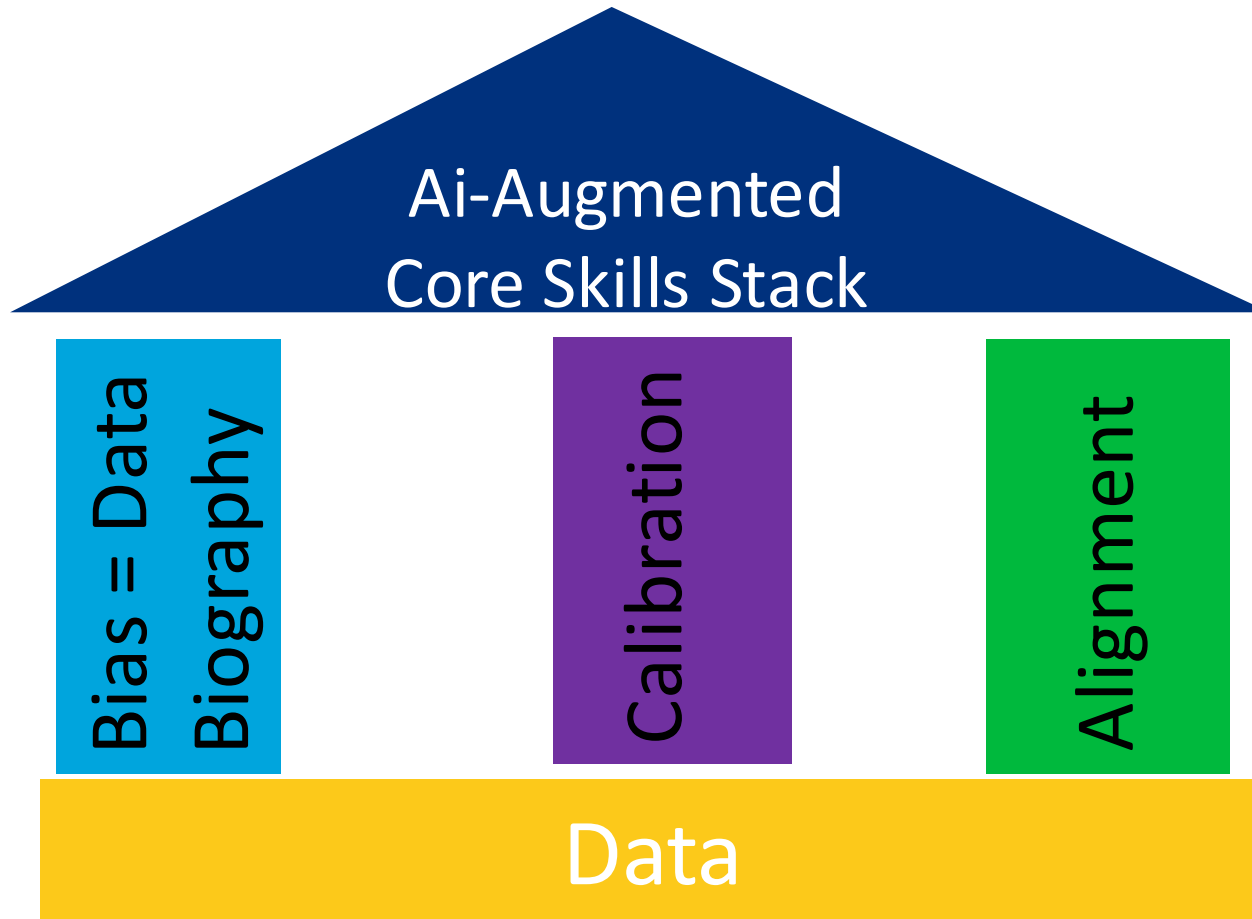
What are we trying to achieve?

How do we get there?

Loss function

Synthesis of the day's material

1. Ai isn't 'new'; it is mostly ML
2. Stacked logistic regressions approximates neural networks
3. Ai systems = Data x (**Representation + Loss function**)x **Optimization**
(ex. gradient descent)
4. Ai systems behave according to what they are rewarded for
(demonstrated by their loss function) **not what we wish they did**



$$\text{Ai} = \text{Data} \times (\text{Representation} \Rightarrow \text{Loss}) \times \text{Optimization}$$

**Think about questions you have about
today's material.**

We will see you tomorrow at 1:00 pm!